SPACE CHALLENGES PROGRAM 2026

TECHNICAL DESIGN DOCUMENT

Armonia

A Three-Axis Reaction-Wheel Balancing Cube

PROJECT TEAM

Athanasia Nikolova
Andrea Liang
Deyan Nikolaev Vlaev
Niccolo Tonetto
Suvanna Viriyanti Wu
Pablo Urioste

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1 Introduction

1.1 Background & Problem Statement

The Cubli is a reaction-wheel inverted pendulum: a cube-shaped platform that balances on an edge or a corner using only internal actuation, with no external supports or contact forces beyond the single line or point of contact with the ground. The concept was introduced by the Institute for Dynamic Systems and Control (IDSC) at ETH Zurich [1, 2], whose original prototype (15 × 15 × 15 cm) mounts one brushless DC motor and flywheel on each of three orthogonal faces. Commanding a change in wheel speed produces, by conservation of angular momentum, an equal and opposite reaction torque on the cube body; braking a wheel abruptly transfers its stored angular momentum to the body in a near-impulsive event, which is what allows the cube to *jump up* from resting on a face into a balance on an edge or corner.

Dynamically, the system is an *underactuated*, nonlinear, unstable-equilibrium control problem: three actuators (the wheels) must stabilise a body with more degrees of freedom than actuated inputs, about equilibria (edge and corner balance) that are open-loop unstable and increasingly coupled across axes as the contact geometry shrinks from a line (edge) to a point (corner). This is structurally the same problem as the classical inverted pendulum, extended from a single rotational axis to a fully three-dimensional, momentum-exchange-actuated body — which is why the Cubli has become a standard benchmark platform in the controls community over the past decade.

The problem this project addresses is therefore twofold: (i) derive and validate a dynamic model and controller capable of stabilising this underactuated system about its unstable equilibria and driving it through commanded reorientations, and (ii) realise this in hardware as a modular, largely parametric design, since final actuator and sensor specifications are not yet confirmed at the time of writing. The target capability set, in order of increasing difficulty, is edge balance, corner balance, controlled rotation, and walking (Section 1.3).

No single subsystem is exotic, but each sits close to a physical bound, and the bounds are coupled: relieving one tightens another. Four constraints drive the design, and sizing is therefore treated as the first engineering activity rather than a detail settled once parts arrive.

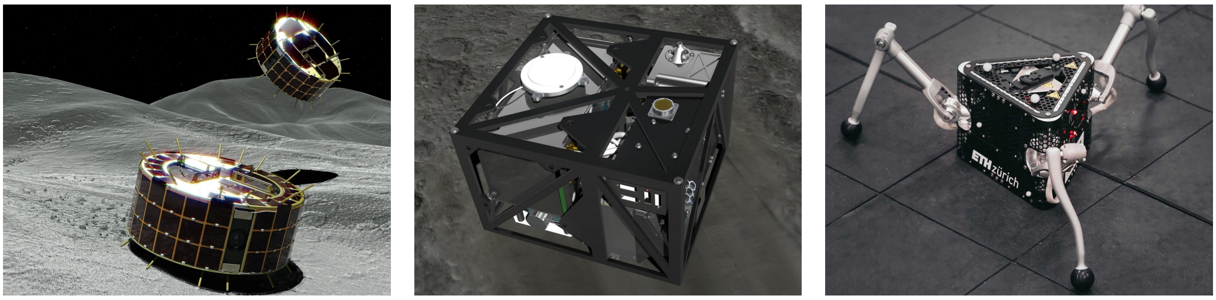
- **The jump-up has no energy margin.** The required per-wheel inertia scales as $ML^{3/2}$ (Section 4.3.1), so mass added anywhere demands a larger wheel, and the wheel is itself part of M. Sizing is a fixed-point iteration, not a one-pass calculation.
- **Braking torque fights the structure.** Momentum accumulated over seconds of spin-up is shed in tens of milliseconds, through one shaft, hub and motor mount — the parts the mass budget wants thin. The structural load case is set by the manoeuvre that proves the machine works.
- **Sensing, not the control law, sets the balancing limit.** Gyroscope bias drifts, and the accelerometer — the only absolute tilt reference — is corrupted by vibration the controller itself generates. Wheel balance is therefore an estimation requirement, not a mechanical nicety.
- **The plant is nonlinear and constrained.** Corner balancing does not decouple into three single-axis problems: the axes are coupled through the contact point and the inertia tensor. The actuators saturate in both torque and speed, so stored momentum must be actively managed. Linearising about upright is adequate for balancing and inadequate for anything more [3].

1.2 Purpose, Application & Mission Context

The Cubli is not presented here as a balancing novelty but as a terrestrial, gravity-loaded demonstrator of **momentum-exchange attitude control** — the same physical mechanism that governs how real spacecraft point themselves. Nearly every three-axis-stabilised satellite holds and changes its attitude using reaction wheels: flywheels whose commanded speed change produces a reaction torque on the spacecraft body, exactly as in the Cubli. Spacecraft typically carry three or four such wheels (often in a pyramidal arrangement for redundancy) — directly analogous to the Cubli's three orthogonal wheels — and this reaction-wheel approach is the standard for CubeSats and small satellites in particular, making the Cubli representative of a real, currently-flown hardware class rather than an idealized toy problem. The trade against control moment gyros is made in Section 3.1.

Framed as an Attitude Determination and Control System (ADCS) problem, the Cubli's core behaviours map directly onto a spacecraft's day-to-day pointing task:

- **Edge / corner balance** → regulation to a commanded reference attitude and disturbance rejection — holding a pointing setpoint against perturbations.
- **Controlled rotation** → a slew manoeuvre: reference tracking to a new commanded attitude, exactly as a satellite re-aims a payload or antenna.
- **Jump / walk** → momentum-exchange hopping mobility, flight-proven on small bodies where wheeled traction is impossible (Figure 1). Walking, as a sequence of controlled directional hops, extends single-hop capabilities and represents an active research frontier (Section 1.3).



(a) MINERVA-II1 (b) MASCOT (c) SpaceHopper

Figure 1: Momentum-exchange actuation in flight and research hardware: (a) JAXA's MINERVA-II1 rovers [5] and (b) the MASCOT lander [6], deployed by Hayabusa2 onto asteroid Ryugu (2018), hop across the surface and self-right by spinning up and braking an internal flywheel or eccentric mass; (c) ETH Zurich's SpaceHopper [7] extends this principle to controlled, directional hopping in low gravity.

The subsystem exercised here is therefore the one a spacecraft uses for pointing: momentum-exchange attitude control under actuator saturation, with desaturation, on an underactuated body whose inertia is imperfectly known. Under gravity the destabilising torque never relents, so an inadequate controller falls over immediately and unambiguously — the demonstration is self-evaluating, which a free-fall attitude testbed is not.

The nominal Cubli problem was solved a decade ago on precisely manufactured hardware [1, 2, 3]; reproducing it is an engineering exercise, and this document claims no more. A commodity build is adopted for three reasons. It can be crashed repeatedly, so the controller can be tested at its limits; its loose tolerances, imperfectly balanced wheels and drifting IMU supply the parameter uncertainty that robust control methods are intended to tolerate; and a fixed motor, print volume and budget constrain the sizing, since a marginal design cannot be rescued by specifying a larger component. The last of these motivates the adjustable ballast scheme of Section 5.3.4.

1.3 Objectives

The project targets four behaviours, deliberately ordered as a difficulty ramp in which each milestone is a control precondition for the next:

1. **Edge balance** — stabilise the cube on a single edge, the least unstable equilibrium (contact is a line, one primary axis of instability).
2. **Corner balance** — stabilise the cube on a single vertex, the fully unstable equilibrium (point contact, all three axes coupled and unstable simultaneously).
3. **Controlled rotation** — execute a commanded slew between equilibria (e.g. edge to edge, or in-place reorientation), demonstrating reference tracking rather than pure regulation.
4. **Walking** — chain repeated, directional controlled falls between edges/corners into a net translation across a surface, re-stabilising after each transition.

These map one-to-one onto the applications of Section 1.2: (1)–(2) demonstrate pointing regulation and disturbance rejection of increasing difficulty; (3) demonstrates slew/reference-tracking; (4) demonstrates momentum-exchange mobility of the kind flown on Hayabusa2. Milestones (1)–(3) are treated as committed deliverables for this design; walking (4) is scoped as the stretch objective, contingent on hardware confirmation and the jump-up feasibility check (angular momentum required to tip the cube vs. available motor/wheel torque), to be resolved once final components are known.

2 Project Organisation

2.1 Team Organisation

The project is developed by a multidisciplinary team, with responsibilities distributed across the main technical domains of the Cubli: mathematical modelling, control, embedded software, mechanical and three-dimensional design, electrical build, and system integration and validation. Table 1 states each member's role and the scope they are accountable for.

Table 1: Team members, contact details and primary roles.

Member	**Contact**	**Role and accountable scope**
Pablo Urioste Alarcón	pb.urioste4@gmail.com +34 654 491 174	**Project management and hardware-facing software.** Schedule, gate criteria and task allocation; motor-driver configuration, the flight-computer–driver link, the sensor drivers and their calibration, and the fixed-rate loop that carries them. Owns the hardware–software seam. Assembles and edits this document.
Deyan Nikolaev Vlaev	vlaev.dido@gmail.com +359 877 082 499	**Dynamics and technical documentation.** Equations of motion, linearisation and jump-up energy sizing; drafting and deepening of this document against the model.
Niccolo Tonetto	niccolo.tonetto@gmail.com	**Control stack and build support.** Single owner of the control and estimation software — estimator, safety and mode logic, and the consolidated control release — written against the sensor interface Pablo maintains. Fabricates the distribution board and cube harness; leads the sacrificial brake and jump-up scope.
Andrea Liang	andrea.liang.2006@gmail.com +31 06 5796 2976	**Plant identification and control design.** Sizing calculations and the inertia and control budgets; sole owner of rig parameter measurement, from the instrumented measurement through to the validated plant model the balancing controllers are designed on.
Suvanna Viriyanti Wu	Viriyanti.wu@gmail.com +39 329 074 7550	**1-DoF model and mechanical build.** Builds and iterates the single-axis prototype and its testbench, fabricates the reaction wheels, and supports the 3D design work.
Athanasia Nikolova	nasia.nikolova@mtel.gr	**3D design and commercialisation.** 3D CAD, frame design and mechanical enclosures; monitors the design against the frozen mass and interface constraints; prototype assembly and the commercial pitch.

2.2 Work Breakdown Structure

This plan runs from project start to the final presentation on 20 August 2026. The work is grouped into five streams — investigation, requirements and documentation; control and software; hardware design and build; three-dimensional design; and electrical and electronics — led by the owners assigned in Table 1. The investigation is kept to the essentials needed to substantiate the design: a focused literature review, the dynamics derivation, and the wheel sizing that follows from it. Table 2 breaks the streams into individual tasks, each with its scope, single accountable lead and dates.

Table 2: Work breakdown structure, broken into individual tasks with scope, lead and dates, grouped by workstream.

ID	Task	Scope	Lead	Dates
Investigation, Requirements & Documentation				
A1	Literature & state of the art	Review the ETH Cubli, the one-wheel Cubli, CubeSat ADCS practice and the hopping-mission precedents	Suvanna	23–25 Jul
A2	Problem statement & framing	Lock the ADCS application narrative that the rest of the document refers back to	Pablo	23–24 Jul
A3	Scope & requirements	Define scope and build the requirements table (functional, performance, design constraint) with verification methods	Pablo	24–25 Jul
A4	Milestones & roles	Prototype staging, binary gate criteria, and a one-page RACI of owners	Pablo	23–25 Jul
A5	Risk register	Trigger, impact, owner and mitigation for the leading risks	Pablo	28–29 Jul
A6	1-DoF equations of motion	Lagrangian derivation of the reaction-wheel pendulum with friction terms	Deyan	24–25 Jul
A7	3D equations of motion	Coupled three-wheel dynamics for the edge and corner pivots	Deyan	27–28 Jul
A8	Linearisation & controllability	Equilibria, linear models, controllability/observability checks	Deyan	29–31 Jul
A9	Jump-up sizing & budget	Energy-method wheel sizing and the parametric mass/inertia/CoM budget	Deyan	30 Jul–2 Aug
A10	TDD draft (all sections)	Every member drafts their own section; placeholder figures acceptable	Deyan	25–28 Jul
A11	TDD deepen + rig evidence	Modelling and control sections in full, with the rig evidence and measured parameters available at the time of writing	Deyan	29 Jul–2 Aug
A12	TDD v1 assemble & submit	Consistency pass, figures, references, submission of the first assessed deliverable	Pablo	3 Aug
A13	Assessor feedback triage	Triage the v1 assessment, assign each comment to an owner, and re-baseline roles and task allocation for v2	Pablo	4–7 Aug
A14	TDD v2 corrections & second draft	Address the assessor comments and fold in the build, bring-up and test results since v1. Ends 12 Aug: no work 13 Aug	Pablo	8–12 Aug
A15	TDD v2 final submission	Final consistency pass and submission of the corrected document	Pablo	14 Aug
Control & Software				
B1	Sim environment + plant	Simulation setup and the 1-DoF plant, validated by energy conservation	Andrea	23 Jul–3 Aug
B2	LQR design (1-DoF, sim)	Tune the regulator and sweep robustness to inertia error, delay and saturation	Andrea	26 Jul–6 Aug
B3	State estimator	Complementary tilt filter, tested against injected noise and bias	Niccolo	27 Jul–7 Aug
B3b	Software interface contract	Fix the sensor and command structs passed between the sensor loop and the controller, with the units and sign convention of every field	Pablo	3–4 Aug

ID	Task	Scope	Lead	Dates
B4	Flight computer & comms	Teensy 4.1 architecture, loop rates, driver configuration and the CAN-FD command/reply protocol	Pablo	24 Jul–7 Aug
B5	Sensor drivers + cal	IMU and encoder drivers, gyro/accel calibration and the lever-arm correction	Pablo	28 Jul–7 Aug
B6	Safety & mode logic	Watchdog, tilt-limit disarm, wheel-speed cap and the mode state machine	Niccolo	30 Jul–7 Aug
B7	1-DoF control design complete	Consolidated 1-DoF control stack — plant, regulator, estimator and safety logic — released as the rig baseline	Niccolo	8–9 Aug
B8	System identification	Measure K_t, wheel inertia, friction and loop latency on the rig. Single owner with E6	Andrea	7–9 Aug
B9	3D model + LQR (sim)	3D plant and the edge/corner regulators with the CAD inertia tensor	Andrea	10–14 Aug
B10	EKF / MEKF estimator	3D estimator with gyro-bias states; characterise the yaw drift	Niccolo	12–15 Aug
B11	Edge balance (S2a)	Retune the edge controller on the assembled cube; log disturbance recovery	Andrea	15–17 Aug
B12	Corner balance + slew (S2b/c)	Coupled three-axis corner balance and a commanded slew; report ADCS metrics	Andrea	17–19 Aug
B13	Jump-up (S3)	Spin-up profile and brake release for a face-to-edge jump. Sacrificial scope: yields to G6 and G7 if the schedule is contested	Niccolo	19–20 Aug
Hardware Design & Build				
C1	Envelope & layout study	Packaging of wheels, drivers, battery and electronics with a central CoM	Nasia	23–24 Jul
C2	Reaction-wheel design	Wheel to the inertia target: radius, rim, hub bolting to the motor bell	Suvanna	24–26 Jul
C3	Mounts & brackets	Motor mount, ring-magnet carrier and adjustable encoder bracket	Nasia	27–29 Jul
C4	1-DoF rig CAD + jigs	Single-axis rig and the balancing/assembly tooling	Suvanna	25–28 Jul
C6	Procurement order (round 1)	Order the long-lead items (rims, frame stock, connectors, spares)	Pablo	23–24 Jul
C7	Bench PSU & bring-up	Current-limited supply, safety kit, and single-axis electrical bring-up	Pablo	25–27 Jul
C8	Fabricate wheel rings	Print the PET-CF rings and hubs to the frozen wheel geometry	Suvanna	27 Jul–2 Aug
C8b	Wheel ballast & completion	Install the M6 bolt/nut ballast stations to the sizing specification and complete each wheel	Suvanna	4–6 Aug
C9	Balance & spin-test	Static-balance and containment spin-test every wheel, including a per-wheel ballast-count verification	Suvanna	5–6 Aug
C10	Assemble 1-DoF rig	Motor, balanced wheel, encoder gap, IMU and hard stops, on reinforced rig supports	Pablo	31 Jul–2 Aug
C11	Print structure + frame	Structural print set and the fabricated, orthogonal frame	Suvanna	8–12 Aug

ID	Task	Scope	Lead	Dates
C12	Perfboard + harness	Power/signal board and the full power and CAN harness	Pablo	8–12 Aug
C13	Cube integration + power-on	Assemble the three axes, then bench and battery smoke test	Pablo	13–15 Aug
C14	Brake mechanism build	Fabricate and fit the three servo brakes; bench-test the deceleration	Suvanna	15–18 Aug
3D Design				
D1	Concept sizing & fit study	Confirm that every component fits the 150 mm envelope and lay out the internal structure	Nasia	27 Jul–2 Aug
D2	Frame design freeze	Freeze the inertia-bearing frame geometry. Later features that do not affect the mass properties may still be added	Nasia	2 Aug
D3	Mass properties & budget closure	Export the inertia tensor and CoM and close the mass budget against the sizing analysis	Nasia	3–4 Aug
D4	Detail CAD	Brackets, electronics bay, harness routing and the electrical keep-outs to the frozen board outline	Suvanna	3–5 Aug
D5	Procurement order (round 2)	Consolidate the bill of materials arising from the 3D design and place the second order	Pablo	5 Aug
D6	Tolerance & clearance analysis	Fit and tolerance study, including the wheel sweep envelope and fastener access	Nasia	5–6 Aug
D7	Manufacturing package release	Issue the drawings, print set and assembly instructions for fabrication	Nasia	7 Aug
Electrical & Electronics				
E1	Wiring & placement design	Electrical architecture, wiring scheme and component placement for the single-axis rig, issued as the build reference	Pablo	28–31 Jul
E2	Rail, driver & encoder bring-up	5 V rail soldered and trimmed to 5.00 V under load; driver phase-wired and FOC-calibrated; MA600 encoder mounted at $\leq$ 0.5 mm air gap with verified read-back. Open-loop operation on the bench supply	Pablo	1–3 Aug
E3	Rig bus & sensor integration	Two-node CAN-FD bus with split termination at both physical ends, Teensy and IMU integrated on the rig harness	Pablo	4–5 Aug
E4	Board outline & floorplan	Driver placement from the encoder cable limit, CAN routing, and the perfboard floorplan issued as a frozen outline to CAD	Pablo	4–5 Aug
E5	1-DoF closed-loop bring-up	Closed loop on the rig with the disarm path and watchdog verified before any balance attempt	Pablo	6–7 Aug
E6	Rig parameter measurement	Instrumented measurement of K_t, wheel inertia, friction and loop latency. Merged with B8: one owner measures and models	Andrea	7–9 Aug

ID	Task	Scope	Lead	Dates
E6b	Mechanical & performance validation (1-DoF)	Reconcile the measured torque, inertia, friction and achievable momentum against the jump-up sizing (A9). Deviation beyond tolerance forces a re-sizing decision that day, while the wheel geometry can still change	Andrea	9 Aug
E7	Perfboard + battery power chain	Populate the distribution perfboard (two ground domains, one star point, split 5 V rail); XT60 chain with anti-spark, fuse and ≥ 50 V bulk capacitance; replicate the remaining actuation sets	Niccolo	8–11 Aug
E8	4-node bus + cube harness	Bus scaled to four nodes with stubs < 10 cm and an error soak at 5 Mbit/s; full power and signal harness fabricated and routed; Wi-Fi telemetry for untethered operation	Niccolo	11–12 Aug
E9	Cube power-on, 48 A + thermal	First battery power-on and the three-axis 48 A transient; driver thermals in the enclosed volume; all axes addressable	Pablo	14–15 Aug
E10	Brake servo rail + measurement	Servo on an isolated 5 V rail with PWM drive and flyback; rail stability under stall, brake closure time and spin-down brake torque measured. Sacrificial with C14/B13	Niccolo	16–19 Aug
Deliverables				
F1	Results, demo & final presentation	Slow-motion capture, results plots, deck, and the final presentation	Suvanna	18–20 Aug

2.3 Schedule and Contingency Plan

The full-page schedule in Figure 2 places every task of Section 2.2 on the 23 July–20 August 2026 calendar, and Table 3 states the gate that closes each phase.

Structure of the plan. The first two weeks establish the analytical and design foundation: the dynamics derivation and wheel sizing on the documentation side, the single-axis rig and its wiring design on the hardware and electrical sides, and the simulation environment on the control side. These converge on 2 August, which carries two objectives: the first powered bring-up of the single-axis rig, taking the 5 V rail, the driver and the encoder to verified open-loop operation (M1); and the freeze of the three-dimensional frame geometry, which closes the mass and inertia budget against the sizing analysis (M2). The frame freeze releases the second procurement round on 5 August (M3), which is the last point at which parts can be ordered with useful delivery margin before cube assembly begins on 13 August.

The documentation stream carries two assessed submissions. The first is delivered on 3 August (G4) with the evidence available at that date; the assessor's comments are triaged during the following week and the corrected second draft is submitted on 14 August (M8), by which point the rig results and the integrated cube can both be reported.

Critical path and contingency. The critical path runs from the 2 August bring-up through the closed-loop rig (M4), the measured plant parameters (M6) and the single-axis control design (M7), into the three-dimensional controller and the balancing demonstrations. The single-axis rig is the protected item, since its balance result is the experimental evidence in the document. The three-dimensional design proceeds in parallel on the mechanical side, so that fabrication is not held by the control work.

The plan carries the full scope through to the jump-up demonstration. The final week is nonetheless tightly constrained: edge balance begins two days after cube assembly completes, and the jump-up gate falls on the day of the final presentation. The jump-up (S3) is therefore designated sacrificial scope by standing decision. If the balancing demonstrations require the time, the jump-up yields first, and the demonstration presented on 20 August is the corner balance and commanded slew. The brake mechanism build and its associated electrical work are sequenced after the edge-balance gate for the same reason.

Table 3: Milestone gates, from kick-off to the final presentation, with the criterion that closes each one.

Gate	Pass criterion	Date
G1	Long-lead order placed; requirements and plan locked	24 Jul
G2	Dynamics, linearisation and wheel sizing complete	31 Jul
M1	1-DoF rig bring-up: 5 V rail trimmed under load, driver commutating, encoder verified at ≤ 0.5 mm gap	2 Aug
M2	3D frame geometry frozen at 150 mm; mass, inertia and CoM budget closed against the sizing analysis	2 Aug
G4	TDD v1 delivered for assessment	3 Aug
M3	Procurement round 2 consolidated and ordered	5 Aug
M4	1-DoF closed loop live; disarm path and watchdog verified	7 Aug
M5	Manufacturing package released for fabrication	7 Aug
M6	Plant parameters measured on the rig and validated against simulation; measured torque, inertia and friction reconciled with the sizing analysis, or a re-sizing decision recorded	9 Aug
M7	1-DoF control design complete: plant, regulator, estimator and safety logic	9 Aug
G3	1-DoF rig balances $\geq$60 s unassisted	12 Aug
M8	TDD v2 corrected and finally submitted	14 Aug
G5	Cube integrated and powered; all axes addressable	15 Aug
G6	Edge balance (S2a) for $\geq$60 s with disturbance recovery	17 Aug
G7	Corner balance and commanded slew (S2b/c)	19 Aug
G8	Face-to-edge jump-up (S3) captured on video (sacrificial scope)	20 Aug
G9	Final presentation and demonstration	20 Aug

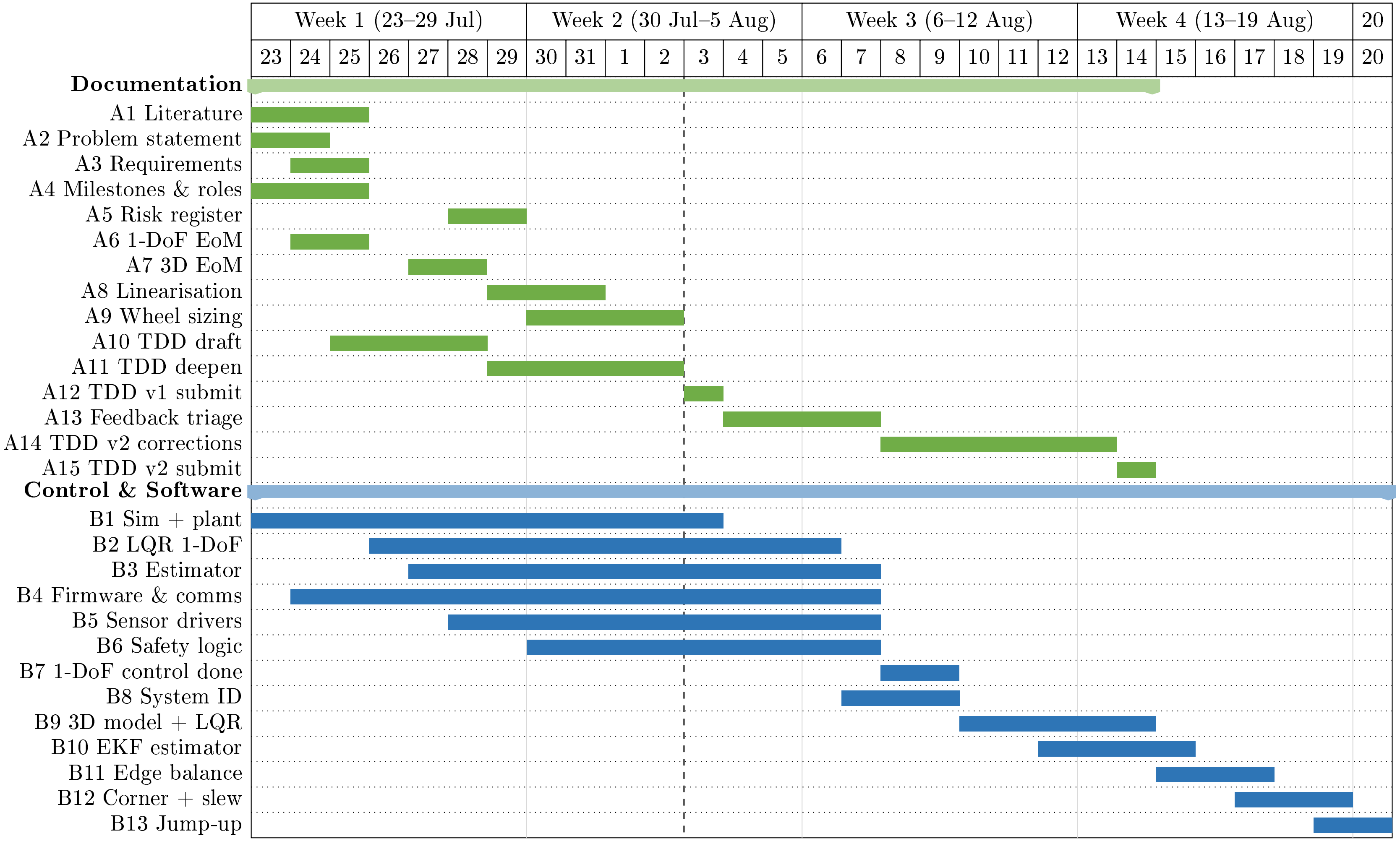


Figure 2: Project schedule, 23 July–20 August 2026 (sheet 1 of 3): documentation and control/software streams. The documentation stream (green) carries two assessed submissions, on 3 and 14 August. Control and software (blue) runs from the simulation environment through to the balancing demonstrations, with the three-dimensional extension concentrated in the final two weeks. The dashed vertical rule marks the current date at the time of compilation. The build streams continue in Figure 3 and the milestone gates in Figure 4.

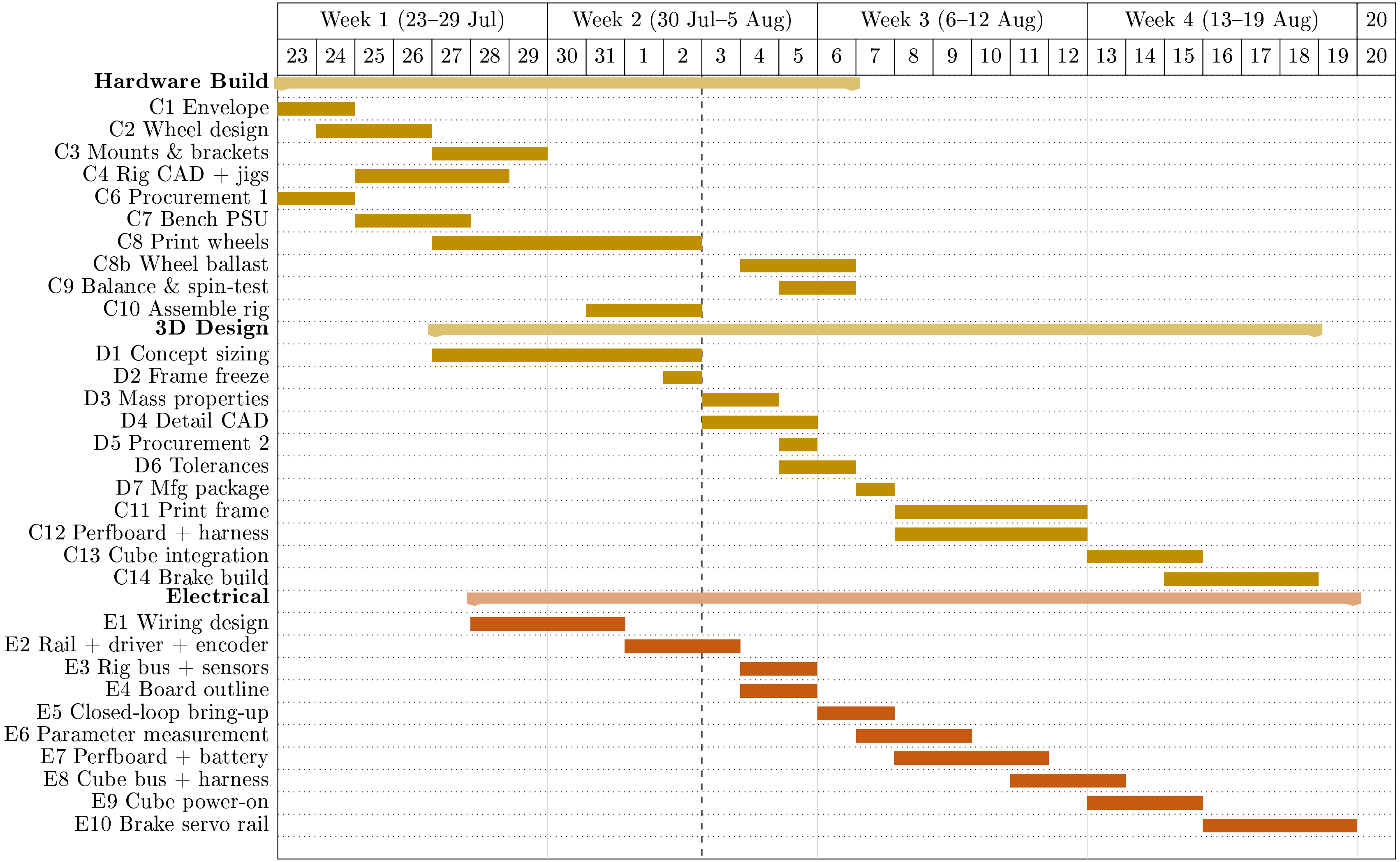


Figure 3: Project schedule, 23 July–20 August 2026 (sheet 2 of 3): hardware, three-dimensional design, electrical and deliverable streams, continued from Figure 2. Hardware (gold) separates the single-axis build from the three-dimensional design, and the electrical stream (orange) proceeds from the single-axis bring-up to the integrated cube. The dashed vertical rule marks the current date at the time of compilation. The milestone gates that close each phase are shown in Figure 4.

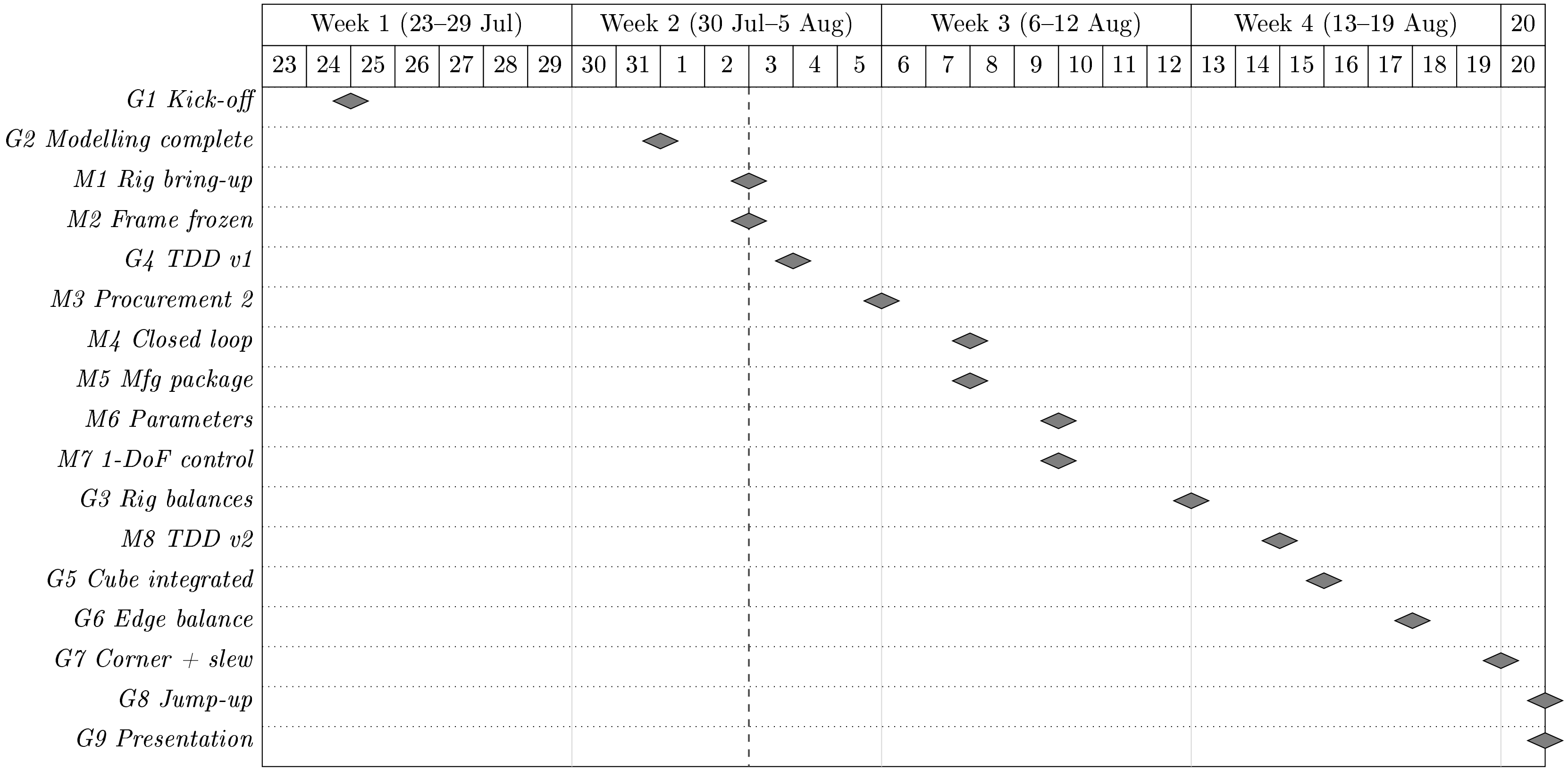


Figure 4: Project schedule, 23 July–20 August 2026 (sheet 3 of 3): the milestone gates that close each phase, continued from Figures 2 and 3. Grey diamonds mark the gates listed in Table 3, placed on the day each gate falls due. The dashed vertical rule marks the current date at the time of compilation. All three sheets share the same time axis, so columns align between them.

3 System Analysis & Requirements

3.1 Justification of reaction wheels for the ACDS system

Since this is a proof-of-concept demonstration of the control loop, a system actuated with orthogonal reaction wheels is well-suited to represent principles applicable to smallsat ADCS. The momentum-exchange actuation principle and closed-loop control architecture are the same as those used on real spacecraft, even though the Cubli's dynamics (balancing against gravity) differ from a satellite's free-fall attitude control problem. Reaction wheels are also structurally simpler and require less specialized hardware than control moment gyros, which offer higher torque density but at the cost of gimbal mechanisms and more complex control software, complexity that is typically only justified on larger spacecraft. For a low-cost, resource-limited project, a reaction wheel Cubli therefore demonstrates the core actuation and control principles of smallsat ADCS in a practical and relevant way.

3.2 Requirements & Objectives Traceability

Each project objective (Section 1.3) imposes a required capability, which in turn is confirmed by a specific test in the validation plan . The mapping below keeps every requirement traceable from motivation to verification.

Table 4: Objective-to-requirement-to-verification traceability.

Objective	Required capability	Verified by
Edge balance	Regulation about a line-contact equilibrium	Sec. **??**
Corner balance	Regulation about a point-contact equilibrium, all axes	Sec. **??**
Controlled rotation	Reference tracking / slew between equilibria	Sec. **??**
Walking (stretch)	Chained jump-up + re-stabilisation	Sec. **??**

3.3 Coordinate Frames & Conventions

4 Dynamic Modelling & Control

The full Cubli is underactuated and simultaneously unstable about all three axes when balanced on a corner (Section 1.3). The modelling, identification, and control workflow is therefore first validated on a single-face 1-DoF prototype, whose planar dynamics are the two-dimensional (2D) model used throughout this section, before being generalized to the 3D cube in Section 4.5.

4.1 2D Reaction-Wheel Pendulum

The 1-DoF prototype is a 3D-printed plate sized to one cube face, with a motor-driven momentum wheel at its centre and a braking mechanism at one corner. The plate pivots on a bearing at that corner, constraining motion to a single rotational degree of freedom in a vertical plane.

4.1.1 Equations of Motion

The nonlinear equations of motion are

$$\ddot{\theta}_b = \frac{(m_b l_b + m_w l)\, g \sin\theta_b - T_m - C_b\dot{\theta}_b + C_w\dot{\theta}_w}{I_b + m_w l^2}, \tag{1}$$

$$\ddot{\theta}_w = \frac{(I_b + I_w + m_w l^2)(T_m - C_w\dot{\theta}_w)}{I_w(I_b + m_w l^2)} - \frac{(m_b l_b + m_w l)\, g \sin\theta_b - C_b\dot{\theta}_b}{I_b + m_w l^2}, \tag{2}$$

with motor current-torque relation $T_m = K_m u$. Symbols are defined in Table 5 and used throughout this section.

Table 5: 2D pendulum model variables.

Symbol	Description	Unit
θ_b	Body tilt angle	rad
θ_w	Wheel rotation relative to body	rad
m_b, m_w	Body, wheel mass	kg
I_b, I_w	Body (about pivot), wheel (about motor axis) inertia	$\mathrm{kg\,m^2}$
l	Motor axis to pivot distance	m
l_b	Body CoM to pivot distance	m
g	Gravitational acceleration	$\mathrm{m/s^2}$
T_m	Motor torque	N m
C_b, C_w	Body, wheel damping coefficient	$\mathrm{kg\,m^2/s}$
K_m	Motor torque constant	N m/A
u	Motor current input	A

4.2 Parameter Identification & the 2D Testbed

CAD mass-properties simulation, accounting for print material, infill, and mounted hardware (driver, encoder, MCU, gyroscope/accelerometer, wiring), gives m_b, m_w, I_b, I_w directly, while l and l_b follow from the plate's design geometry. The damping coefficients C_b and C_w cannot be obtained from CAD, since they depend on bearing friction, wiring drag, and motor cogging; nominal values will be estimated for the initial control design, with dedicated testing planned once the prototype is fabricated.

Table 6: Parameter determination method (symbols per Table 5).

Symbol	Determination method
l	Calculated during design
l_b	Calculated during design
m_b	CAD mass properties
m_w	CAD mass properties
I_b	CAD mass properties
I_w	CAD mass properties
C_b	Estimated (future testing planned with prototype)
C_w	Estimated (future testing planned with prototype)
K_m	Motor datasheet

4.3 Jump-Up Feasibility Analysis

Given CAD-derived $I_b, I_w, m_b, l_b, m_w, l$, the wheel speed ω_w required for jump-up is estimated before fabrication, assuming a perfectly inelastic wheel-body collision. Conservation of angular momentum at impact and of energy from impact to edge-standing yield

$$\omega_w^2 = (2 - \sqrt{2})\,\frac{I_w + I_b + m_w l^2}{I_w^2}\,(m_b l_b + m_w l)\,g. \tag{3}$$

This value is checked against the motor's achievable speed at the available bus voltage and used to size wheel inertia (radius, rim mass) within the motor's operating range without excessively reducing the balancing recovery angle. Post-fabrication, ω_w is refined experimentally to correct for deviations from the ideal collision assumption.

The braking mechanism uses a servo-driven barrier that strikes a bolt head on the wheel to stop it abruptly. Servo response time bounds the validity of the "instantaneous stop" assumption; a slower stop yields a lower effective post-impact body velocity than predicted, compensated through the same tuning process.

4.3.1 Torque-Limited Wheel-Inertia Target

The relation above assumes an instantaneous, lossless momentum transfer. For preliminary sizing, the achievable motor speed is treated as fixed and the required per-wheel inertia is derived from a momentum budget that additionally accounts for finite brake torque. This gives an ideal per-wheel angular momentum

$$h_{w,\text{ideal}} = \sqrt{\eta}\,\frac{\sqrt{2g\,\lambda\,\kappa}}{n}\,ML^{3/2}, \tag{4}$$

where M, L, g are cube mass, edge length, and gravity, η is an energy margin, and κ, λ, n are dimensionless geometric factors set by the contact case (Table 7). The corner case is more demanding and governs the design.

Table 7: Contact-case geometric factors.

Case	κ (pivot inertia)	λ (CoM rise)	n (momentum transfer)
Edge	2/3	$1/\sqrt{2} - 1/2$	1
Corner (governs design)	11/12	$\sqrt{3}/2 - 1/2$	$\sqrt{2}$

A finite brake torque τ_b cannot deliver the impulse instantaneously and must exceed the gravitational tip-over torque $\tau_g = MgL/2$. This loss is captured by a brake-amplification factor

$$\beta = \frac{\tau_b}{\tau_b - \tau_g}, \qquad h_w = \beta\, h_{w,\text{ideal}}, \qquad I_{w,\text{target}} = \frac{h_w}{\omega_{\max}}, \tag{5}$$

where $\omega_{\max}$ is the maximum practical wheel speed. Equations (4) and (5) are the sizing relations used at step 4 of the closure procedure of Section 5.2: given a candidate cube mass and edge length they return the inertia the wheel of Section 5.3.4 must meet. Their inputs are recorded in Table 8 and are not fixed here—M and L are outputs of the packaging closure, not assumptions preceding it.

Two features of these relations govern how the iteration behaves. First, $h_w \propto ML^{3/2}$, so growing the cube to accommodate a larger wheel also raises the requirement; convergence depends on the achievable inertia growing faster ($\sim D_w^2$) than the requirement does. Second, β diverges as $\tau_b \to \tau_g$: a brake whose torque only marginally exceeds the gravitational tip-over torque cannot deliver the required impulse at any wheel speed. The brake torque must therefore be checked against $\tau_g = MgL/2$ on every iteration, and a comfortable ratio τ_b/τ_g maintained rather than a merely positive margin.

Table 8: Torque-limited wheel-inertia target (corner case). Inputs are fixed by the closure procedure of Section 5.2; the table is populated on each iteration. Values are those of the converged iteration (`analysis/SIZING.py`): M is a fixed point of the closure, not an assumption, reproduced by the Stage 6 roll-up to within 0.1 g. τ_b is an assumed value pending measurement, justified below and equivalent to a 52 ms stop at the present h_w.

Symbol	Description	Value
M	Cube mass	1.772 kg
L	Cube edge length	150 mm
η	Energy margin	1.05
τ_b	Brake torque per wheel	5 N m (assumed; see below)
$\tau_g = MgL/2$	Gravitational tip-over floor	1.303 N m
β	Brake-amplification factor	1.353
$\omega_{\max}$	Max wheel speed (design)	6000 rpm*
h_w	Required angular momentum	0.2588 kg m^2/s
$I_{w,\text{target}}$	**Target wheel inertia**	4.119×10^{-4} kg m^2

* The sizing has additionally been run at 5000 rpm as an over-dimensioning check. Since $I_{w,\text{target}} = h_w/\omega_{\max}$, the lower speed raises the inertia requirement by a factor 6/5 and so demands more ballast; a wheel that closes at 5000 rpm therefore closes at 6000 rpm with margin in hand. 6000 rpm is the value the design claims, 5000 rpm the conservative case it was verified against.

Selection of $\omega_{\max}$. The maximum wheel speed is a design choice rather than a datasheet figure, and it enters the sizing directly through $I_{w,\text{target}} = h_w/\omega_{\max}$: a higher assumed speed permits a proportionally smaller wheel. It is therefore the primary relief variable when the envelope caps the wheel below the requirement (step 5, Table 11), but it must be set from what the drive can sustain in service, not from the no-load figure $K_v V_{\text{bus}}$. Three effects separate the two: winding resistance, which costs speed under the torque required to spin the wheel up; the reduced phase voltage available under sinusoidal field-oriented modulation; and the fall of pack voltage through the discharge curve, which must be taken at its lower end if the manoeuvre is to remain available on a partly-depleted pack rather than only on a fresh one. The design value is set below the ceiling those effects imply, leaving margin for motor tolerance, bearing and aerodynamic drag, and the residual uncertainty in M. The resulting back-EMF should remain a comfortable fraction of the usable bus voltage, and the electrical frequency $\omega_{\max} p/2\pi$ at p pole pairs must sit inside the driver's commutation capability—both checks to be confirmed once the motor and pack are selected.

Selection of τ_b. The brake torque requires more care than the other inputs, because the mechanism does not generate it in the way the symbol suggests. The brake is a servo-driven barrier struck by a bolt head on the wheel: the servo only *holds* the barrier in the wheel's path, and the wheel's own momentum does the work of the collision. The servo's stall torque is therefore not τ_b and need not approach it—the TowerPro MG92B of Table 20 is rated near 0.29 N m, roughly one seventeenth of the value used in sizing. The servo must resist the barrier being deflected, not decelerate the wheel.

Because the stop is a collision rather than a friction clutch, τ_b is not an independent property of any component and cannot be read from a datasheet. It is fixed entirely by how quickly the angular momentum is destroyed,

$$\tau_b = \frac{h_w}{\Delta t}, \tag{6}$$

so quoting a brake torque is equivalent to asserting a stopping time. At the present operating point ($h_w \approx 0.19\,\text{kg}\,\text{m}^2/\text{s}$) the design value $\tau_b = 5\,\text{N}\,\text{m}$ corresponds to $\Delta t \approx 38\,\text{ms}$. Stated that way the assumption is open to inspection: 38 ms is characteristic of servo travel rather than of a rigid impact, and a bolt head striking a printed barrier should arrest the wheel in a few milliseconds, implying a substantially larger τ_b. The design value is thus expected to be conservative for *sizing* purposes.

That conservatism is deliberate, and it is asymmetric. Since $\beta = \tau_b/(\tau_b - \tau_g)$ falls towards unity as τ_b grows, underestimating the brake torque inflates the inertia requirement—the safe direction—while overestimating it leaves the wheel undersized. With the present τ_g, raising τ_b from 5–10 N m relaxes $I_{w,\text{target}}$ by only about 12 %, whereas lowering it to 2 N m inflates the requirement by roughly 65 % and to 1.5 N m by over 150 %. The penalty for error is concentrated almost entirely on the low side, and a mid-range value is therefore adopted in preference to the higher figure a short impact would justify.

One consequence must be carried into the structural design rather than left implicit: τ_b is also a *load*. The sizing value is chosen for the dynamics, and the same number should not be used to size the barrier, the bolt, the servo bracket or the wheel spokes, since a genuine few-millisecond stop would put an order of magnitude more torque through that load path. Until measurement settles the question, the structure is to be checked against a short-Δt case and the dynamics against $\tau_b = 5\,\text{N}\,\text{m}$. The measurement that closes this is the spin-down test of Section **??**, which should report both the impulse-equivalent mean $\bar{\tau}_b = h_w/\Delta t$, which is the sizing quantity, and the peak torque, which is the structural one; the two may differ several-fold, and the logging rate must be fast enough to resolve an event of a few milliseconds without smoothing the peak away.

4.4 Control Design on the 2D Model

4.4.1 LQR Baseline Control

Linearising about the upright equilibrium $(\theta_b, \dot{\theta}_b, \dot{\theta}_w) = (0, 0, 0)$ gives

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x := (\theta_b, \dot{\theta}_b, \dot{\theta}_w), \tag{7}$$

with A, B built from the parameters in Table 5. Zero-order-hold discretization at a rate set by the open-loop unstable pole, with a safety margin, gives $x[k+1] = A_d x[k] + B_d u[k]$. A discrete LQR gain $K_d = (K_{d1}, K_{d2}, K_{d3})$ minimises

$$J(u) = \sum_k x^{\mathrm{T}}[k] Q x[k] + u^{\mathrm{T}}[k] R u[k], \tag{8}$$

giving $u[k] = -K_d(\hat{\theta}_b[k], \hat{\dot{\theta}}_b[k], \hat{\dot{\theta}}_w[k])$.

The tilt estimate $\hat{\theta}_b$ is obtained from a gyroscope and accelerometer mounted at, or as close as mechanically feasible to, the pivot point rather than at the plate's centre. This placement is necessary because an accelerometer at radius r from the pivot measures gravity plus the tangential ($r\ddot{\theta}_b$) and centripetal ($r\dot{\theta}_b^2$) acceleration components induced by the plate's rotation, which corrupt the naive estimate $\tan^{-1}(-a_x/a_y)$ during fast motion (e.g., jump-up or aggressive recovery); placing the sensor at $r \approx 0$ makes both terms negligible. Tilt is estimated with a complementary filter combining gyro integration (drift-prone, unaffected by the terms above) with accelerometer inclination (drift-free, but unreliable during shocks):

$$\hat{\theta}_b[k] = (1-\gamma)\big(\hat{\theta}_b[k-1] + \dot{\theta}_{b,\mathrm{gyro}}[k]\,\Delta t\big) + \gamma\,\theta_{b,\mathrm{accel}}[k], \tag{9}$$

where $\theta_{b,\mathrm{accel}}[k] = \tan^{-1}(-a_x[k]/a_y[k])$ is the instantaneous accelerometer-only estimate and $\gamma \in (0,1)$ is a small weight favoring the gyro term between corrections.

A constant tilt-sensor offset d, including residual bias from imperfect pivot placement or sensor misalignment, would otherwise appear as a nonzero steady-state wheel speed rather than a corrected angle. A low-pass filter state

$$x_f[k+1] = (1-\alpha)x_f[k] + \alpha\hat{\theta}_b[k] \tag{10}$$

is subtracted from the tilt feedback,

$$u[k] = -K_d(\hat{\theta}_b[k] - x_f[k],\ \hat{\dot{\theta}}_b[k],\ \hat{\dot{\theta}}_w[k]), \tag{11}$$

so d is absorbed by x_f at steady state, requiring no separate calibration stage. This fixed-gain design is the baseline balancing controller; Table 9 summarizes the control variables.

4.4.2 Gain Scheduling & Mode Transitions

The controller switches from the open-loop jump-up sequence to the LQR law once $|\hat{\theta}_b| < \theta_{\mathrm{th}}$, with hysteresis around θ_{th} to prevent mode chattering. The baseline uses a single fixed gain K_d at the upright equilibrium. If recovery degrades near the edges of the operating range, where the small-angle linearisation weakens, gain scheduling will be added: additional gains $K_d^{(i)}$ computed at several linearisation points, with the controller interpolating between them based on $\hat{\theta}_b$, reusing the same discrete structure and offset filter.

4.5 Generalization to the 3D Cube

The dynamics, identification workflow, and control approach validated on the 2D model extend to the full three-dimensional cube, where three orthogonal reaction wheels must stabilise the body about its edge and corner equilibria.

Table 9: Control-design variables (2D and corner-balancing 3D model).

Symbol	Description
x	State vector
A, B	Continuous-time state/input matrices (linearised model)
A_d, B_d	Discrete-time (ZOH) counterparts of A, B
K_d	LQR feedback gain
Q, R	LQR state/input weighting matrices
$J(u)$	LQR cost function
$\hat{\theta}_b, \hat{\dot{\theta}}_b$	Estimated tilt angle, rate (gyroscope/accelerometer)
$\hat{\dot{\theta}}_w$	Estimated wheel rate (encoder)
a_x, a_y	Raw accelerometer readings (tangential, radial axes)
$\theta_{b,\text{accel}}$	Instantaneous accelerometer-only tilt estimate
$\dot{\theta}_{b,\text{gyro}}$	Raw gyro rate measurement
γ	Complementary-filter weight (accelerometer term)
Δt	Control/estimation loop sample period
x_f, α	Offset-correction filter state, filter coefficient
d	Constant tilt-sensor offset
θ_{th}	Tilt threshold, jump-up $\rightarrow$ balancing switch
$K_d^{(i)}$	Scheduled gain at linearisation point i (extension)

4.5.1 Full-Body Equations of Motion

With $R \in SO(3)$ the cube's orientation and $\omega_b \in \mathbb{R}^3$ its body-frame angular velocity, the attitude kinematics are

$$\dot{R} = R\,[\omega_b]_\times, \tag{12}$$

where $[\cdot]_\times$ is the skew-symmetric cross-product matrix. The assembly's angular momentum about the pivot, expressed in the body frame, generalizes the single-axis momentum term of the 2D model:

$$H = I\omega_b + \sum_{i=1}^{3} I_{w,i}\,\omega_{w,i}\,e_i. \tag{13}$$

Euler's equation gives

$$\dot{H} + \omega_b \times H = \tau_g - C_b\,\omega_b, \tag{14}$$

and each wheel obeys

$$I_{w,i}\big(\dot{\omega}_{b,i} + \dot{\omega}_{w,i}\big) = T_{m,i} - C_{w,i}\,\omega_{w,i}, \qquad i = 1, 2, 3. \tag{15}$$

Table 10 defines all symbols.

Restricting motion to a single vertical plane with two wheel torques set to zero recovers the 2D model exactly, with $I_b, l_b, m_b, m_w, I_w, C_b, C_w$ the corresponding single-axis projections of $I, r_{cm}, m, I_{w,i}, C_{w,i}$.

4.5.2 Edge vs. Corner Equilibria & Linearisation

Jump-up proceeds in two stages: Flat-to-Edge, where one wheel brakes, followed by Edge-to-Corner, where a second wheel brakes. **Edge balancing** constrains the body to a single axis, so

Table 10: Full-body (3D) model variables.

Symbol	Description
R	Cube orientation relative to world frame
ω_b	Body angular velocity, body frame ($\in \mathbb{R}^3$)
$\omega_{b,i} = e_i^{\mathrm{T}} \omega_b$	Body angular velocity about wheel axis i
$\omega_w = (\omega_{w,1}, \omega_{w,2}, \omega_{w,3})$	Wheel angular velocities relative to body
e_i	Spin axis of wheel i (face normal)
I	3×3 inertia tensor, body+wheels, about pivot
H	Total angular momentum about pivot
$\tau_g = m\, r_{cm} \times (R^{\mathrm{T}} \hat{g})$	Gravity torque about pivot
m	Total assembly mass
r_{cm}	Body-frame vector, pivot to assembly CoM
$\hat{g}$	World vertical unit vector
C_b	3×3 body damping matrix
$I_{w,i}, C_{w,i}$	Inertia, damping of wheel i
$T_{m,i} = K_{m,i} u_i$	Motor torque, wheel i
$u_i, K_{m,i}$	Motor current, torque constant, wheel i

the dynamics reduce to the 2D form, and the 2D LQR baseline and offset filter apply directly, with m_b, l_b, I_b replaced by their edge equivalents.

Corner balancing is genuinely 3D: all three wheels contribute across three rotational DOF. Linearising about the corner-up equilibrium $(\theta, \omega_b, \omega_w) = (0, 0, 0)$, with $\theta \in \mathbb{R}^3$ the small orientation error, gives the same state-space form as the 2D case:

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x := (\theta, \omega_b, \omega_w) \in \mathbb{R}^9, \tag{16}$$

with $A \in \mathbb{R}^{9\times 9}$ and $B \in \mathbb{R}^{9\times 3}$ built from I, r_{cm}, and the per-wheel parameters $I_{w,i}, C_{w,i}, K_{m,i}$ (Table 10). The LQR design generalizes directly: K_d becomes a 3×9 gain matrix, using the same discretization, cost function, and offset filter as Table 9, with Q, R weighting all three tilt axes, body rates, and wheel speeds. The mode-transition logic gains a third mode, Edge-to-Corner jump-up, triggered once edge balancing is stable, followed by a second threshold switch into the corner LQR once estimated orientation nears corner-up.

5 Preliminary Hardware Design

This design is preliminary: final actuator and sensor specifications are not yet confirmed, so the mechanical and electrical architecture is kept modular and largely parametric (Section 1.1).

5.1 System Architecture

Figure 5 shows the closed control loop that ties the four subsystems together. The Teensy 4.1 runs estimation and control at 1 kHz and issues torque commands over CAN-FD to three moteus-n1 drivers, each closing a 30 kHz field-oriented current loop on its MN4006 motor. The resulting reaction torque acts on the cube body; the body's attitude is sensed by the BMI270 IMU and the rotor angle by the MA600 absolute encoder, closing the loop back to the controller. The two loops run at different rates by design: the fast current loop is delegated entirely to the drivers, so the flight controller need only meet the 1 ms outer-loop deadline of Section 4.4.

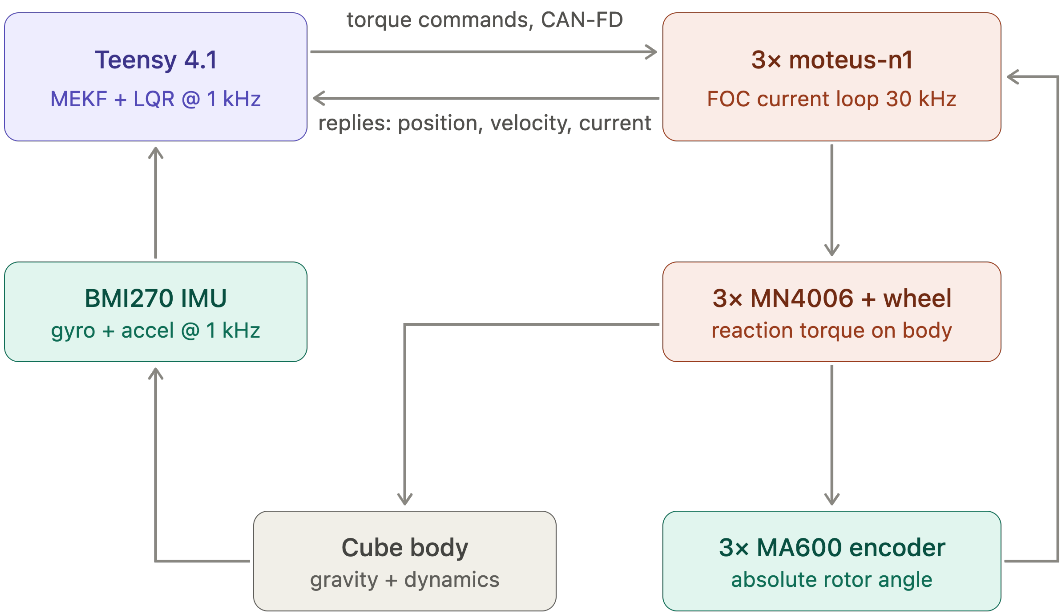


Figure 5: System architecture and control loop. Blue: processing; green: sensing; orange: actuation. Rates shown are the design targets of Section 4.

5.2 Sizing Constraints and Closure Procedure

The cube edge length L and the reaction-wheel inertia I_w are not independent design choices: they are the two ends of a single constraint chain that begins with what must physically be carried inside the cube. This section states that chain and the order in which it is closed. No dimensions are committed here—component selection is still open (Section 5), and the numbers follow from the procedure rather than preceding it.

The chain runs in the order set out in Table 11. The volume claim inside the cube is dominated by the LiPo pack and the inner structure that carries it — a rigid subframe locating the three motors on mutually perpendicular axes through a common centre, together with the drivers, flight controller, IMU and brake servos. The pack is incompressible and sits near the geometric centre so as not to skew the centre of mass off the body diagonal, so L is the smallest edge length containing that volume plus wall thickness and the contact features. L then caps the wheel diameter D_w, and with it the achievable inertia $I_w \sim m_w D_w^2/4$; ballast cannot rescue a wheel that is short at its maximum diameter, because it necessarily sits inboard of the ring it displaces.

Against that stands the requirement from the jump-up analysis (Section 4.3.1),

$$I_{w,\text{target}} = \frac{h_w(M, L)}{\omega_{\max}}, \qquad h_w \propto M L^{3/2},$$

which rises with both M and L. Two consequences close the loop. Enlarging the cube to gain wheel diameter also raises the requirement, but the achievable inertia grows faster than the requirement does (D_w^2 against $L^{3/2}$), so a wheel short at a given L can be closed by a modest increase in L. And any mass added in doing so feeds back through $h_w \propto M$ and through the gravitational tip-over torque $\tau_g = MgL/2$ that the brake must exceed — which is why the procedure terminates at a fixed point rather than in one pass.

Table 11: Sizing closure procedure. Steps are executed in order; a failure at step 4 returns to step 3 or step 1 as indicated.

Step	Determines	Governing consideration
1	Battery pack geometry	Capacity and discharge rating for the jump-up duty cycle
2	Inner clear volume $V_{\min}$, hence minimum L	Pack + motor subframe + driver/controller stack, non-intersecting; pack near centre
3	Maximum wheel diameter D_w, hence achievable I_w	Clearance to subframe, motor body and the two orthogonal modules
4	Requirement $I_{w,\text{target}} = h_w(M, L)/\omega_{\max}$	Corner jump-up, Eqs. (4)–(5)
5	Accept or iterate	If $I_w < I_{w,\text{target}}$: raise L (step 2), raise $\omega_{\max}$, or trim M; re-enter at step 3
6	Mass reconciliation	Updated M feeds back into step 4; iterate to a fixed point

The two contact cases are not equally demanding. The corner equilibrium requires a larger rise of the centre of mass and a less favourable momentum-transfer geometry than the edge equilibrium, so for a common maximum wheel speed $\omega_{\max}$ it demands the greater wheel inertia (Table 7). Step 4 above is therefore always evaluated for the corner case; a wheel that satisfies the corner requirement satisfies the edge requirement with margin. Table 12 records both once the procedure closes.

Table 12: Required per-wheel inertia by contact case, evaluated at $\omega_{\max} = 6000$ rpm for the converged $M = 1.772$ kg. Only the geometry factor differs between the cases: τ_g and β are properties of the cube and the brake, not of the feature it stands on.

Contact case	h_w (kg m^2/s)	$I_{w,\text{target}}$ (kg m^2)	Relative
Edge	0.2348	3.737×10^{-4}	90.7 %
Corner (critical)	0.2588	4.119×10^{-4}	100 %

The corner case demands 9.3 % more inertia than the edge case, so sizing on it satisfies the edge requirement with that margin in hand. The selected wheel delivers 4.599×10^{-4} kg m^2 (Table 15), which is 111.7 % of the corner requirement and 123.1 % of the edge requirement.

The allocation of Table 13 is the bookkeeping side of steps 5 and 6: it is populated as components are selected and re-totalled on each iteration, since the total is an input to $I_{w,\text{target}}$ as well as an output of the design.

Table 13: Mass budget (per cube), rolled up from the bill of materials by `analysis/SIZING.py` (Stage 6). The total is an input to the inertia requirement, not only a result: the value below is the *converged* one, which the closure reproduces to within 0.1 g of the M assumed in Table 8.

Subsystem	Mass (g)	Share	Basis
Reaction wheels (3×172.5 g)	517.5	29.2 %	Sized (Sec. 5.3.4)
Motors and brake servos ($3\times$ each)	245.4	13.9 %	Datasheet
Power (battery, XT60, regulator)	279.0	15.7 %	Drives envelope (step 1)
Electronics (drivers, MCU, encoders, IMU)	74.2	4.2 %	Datasheet
Wiring, perfboard and passives	45.5	2.6 %	Estimated
Subtotal — specified hardware	**1161.6**	65.6 %	Measured or datasheet
Structure (subframe, frame, fasteners)	610.0	34.4 %	**Allowance, pending CAD**
Total M	**1771.6**		**Converged fixed point**

The structure line is an *allowance*, not a measurement, and at 34 % of the total it is the dominant uncertainty in the budget. It is reported separately from the specified hardware for that reason: the subtotal is what the design currently knows, and the total is what it currently projects. Closing the CAD model replaces the allowance with a real figure and re-enters the loop at step 6.

5.3 Mechanical Design and CAD Construction

This subsection states the mechanical concept and the CAD conventions the build follows. It is deliberately kept to the level of decisions and interfaces: the part-by-part model tree, the manufacturing drawings, the fabrication settings and the assembly sequence are recorded in Appendix B, so that a change to a bracket does not require an edit to the design rationale, and the rationale is not buried in geometry.

5.3.1 CAD Methodology and Model Conventions

The model is built top-down from the envelope closure of Section 5.2 rather than bottom-up from individual parts, so that the edge length L and the wheel diameter D_w propagate to every dependent feature when an iteration changes them. Three conventions make that propagation reliable, and they are stated here because they constrain how every part in Appendix B is modelled:

1. **A single skeleton drives the assembly.** L, D_w, the motor axis offsets and the wall thickness live in one parameter table referenced by every part; no downstream part carries a hard-coded dimension that duplicates one of them.

2. **The body frame is the CAD origin.** The assembly origin is the cube's geometric centre, with axes on the three motor spin axes, so that CAD-reported mass properties are directly the I and r_{cm} of Table 10 without a change of basis.

3. **Mass properties are an output of the model, not an annotation.** Every part carries its assigned material and print density, so the assembly mass roll-up is the quantity fed back into Table 13 at step 6 of Table 11.

Convention 3 makes the iteration auditable: the mass budget and the inertia requirement are read from the same model, so a design that closes analytically cannot fail to close in the CAD without the discrepancy being detected.

5.3.2 1-DoF Prototype Build

The single-face rig is the first article built and the bench on which the parameters of Table 6 are measured. Its structure is a 3D-printed plate sized to one cube face, a central motor and wheel mount, a corner bearing defining the single rotational degree of freedom, and encoders and other hardware. Note that the servo (brake system) is not included in the 1-DoF prototype as its purpose is validating the control loop on a more well-understood problem. Detailed geometry and the drawing set are in Appendix B.3.

The rig as built is shown in Figure 6. The plate is mounted diagonally, so that the pivot bearing sits at the lowest corner and the wheel axis passes through the plate centre: this places the centre of mass directly above the contact point at the upright equilibrium, which is the configuration linearised in Section 4.4. The two printed uprights either side of the frame are hard stops, limiting travel to a few degrees past the point of no return so that a failed balancing attempt ends against a stop rather than on the bench.

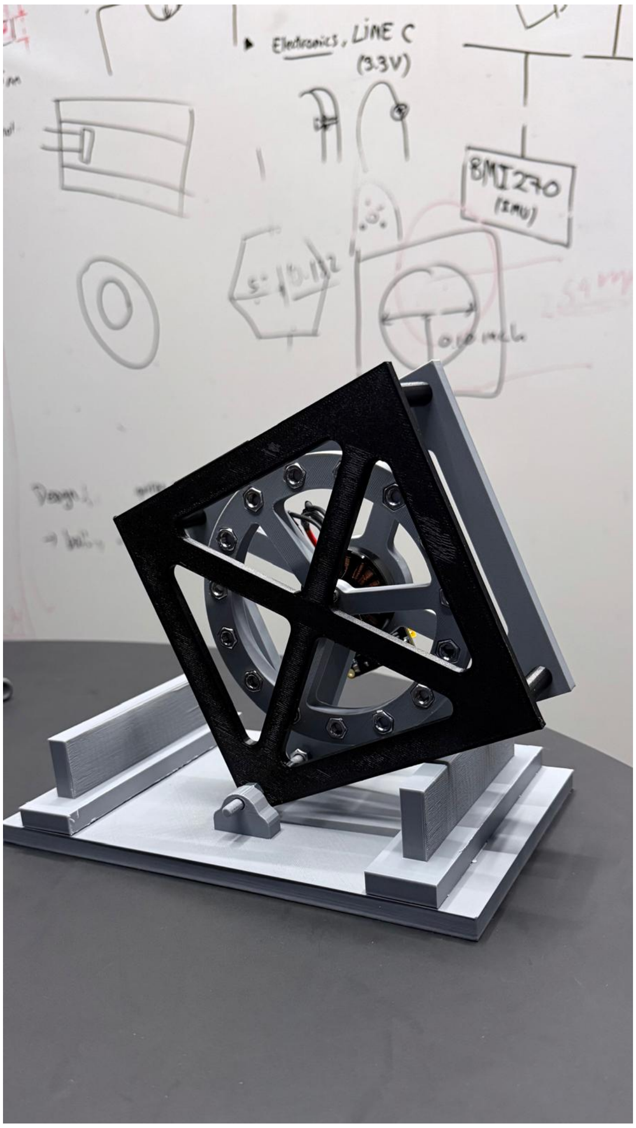


Figure 6: The single-face prototype rig as built: one reaction wheel driven by a brushless motor on a diagonally-mounted printed plate, pivoting on a corner bearing against a printed base with hard stops. This is the article on which the plant parameters of Table 6 are measured and the single-axis controller of Section 4.4 is validated before the three-axis build. The drawing set is given in Appendix B.3.

5.3.3 3D Cube Frame

The frame is conceived in two parts, reflecting the constraint chain of Section 5.2. An *inner structure* locates the three motors on mutually perpendicular axes through a common centre and carries the battery pack, the three drivers, the flight controller, inertial measurement unit, and the brake servos; it is the element that fixes the minimum internal clear volume and therefore the edge length L. An *outer frame* carries the edge and corner contact features, protects the wheels, and reacts the motor mounting loads from the inner structure into the ground contact.

A first physical prototype of this frame has been printed and is shown in Figure 7. It is explicitly a *concept prototype*, not the flight article: it demonstrates the two-part architecture, the X-braced face pattern and the packaging of three orthogonal wheel modules within the cube envelope, and it serves as a physical check on clearances that are awkward to judge on screen. The final design is not frozen. Detailed geometry follows the envelope closure of Section 5.2, and the wheel dimensions in particular remain open pending the sizing iteration of Section 5.3.4 and confirmation of the final motor; the mass properties of the printed prototype are therefore not the mass properties assumed in the budget.

The split is also the main mechanical interface in the machine, and it is managed as one: the inner structure is designed against a fixed bolt pattern and keep-out envelope published to the outer frame, so the two can be revised independently as long as that interface holds. The controlled interfaces are listed in Table 14; the parts realising them, with their drawings, are in Appendix B.4.

Table 14: Controlled mechanical interfaces. Each is frozen once agreed, so that the parts on either side can be revised independently.

Interface	Between	Controlled quantities
Motor mount	Motor ↔ inner structure	Bolt circle, shaft height, concentricity to the wheel axis
Wheel hub	Wheel ↔ motor rotor	Bore, bolt pattern, axial location, balance datum
Frame joint	Inner structure ↔ outer frame	Bolt pattern, keep-out envelope, load path to the contact features
Electronics mount	Board ↔ inner structure	Board outline, mounting-hole pattern, keep-out heights, connector exit directions (Sec. 5.4.5)
Brake mount	Servo ↔ inner structure	Servo location, barrier throw
Battery retention	Pack ↔ inner structure	Pack envelope, retention method, centring tolerance to the cube centre
Contact features	Outer frame ↔ ground	Corner and edge geometry, material, replaceability

5.3.4 Reaction-Wheel Sizing

The wheel must supply the per-wheel inertia $I_{w,\text{target}}$ demanded by the corner jump-up (Section 4.3.1), at an outer diameter no greater than the envelope permits (step 3 of Table 11), while remaining light enough not to dominate the cube mass—since its own mass feeds back into $I_{w,\text{target}}$ through M. These three demands pull against each other, and the wheel architecture is chosen so that the design can be trimmed against them after fabrication rather than having to be right the first time.

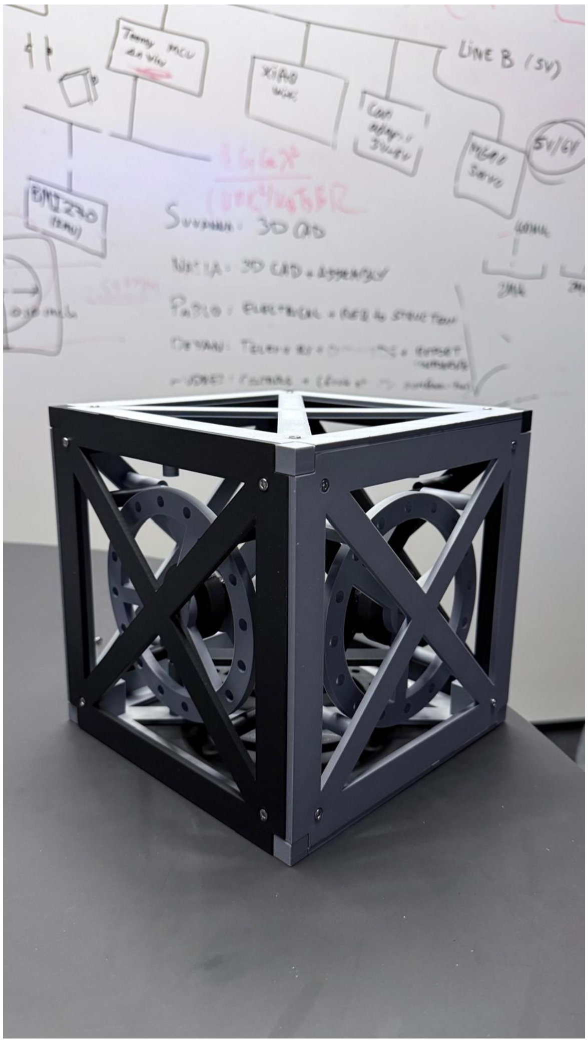


Figure 7: Printed concept prototype of the three-axis cube frame, showing the X-braced outer faces and the three orthogonal reaction-wheel modules carried by the inner structure. This article validates the two-part architecture and the internal packaging only: it is *not* the final design. Wheel geometry and the frame edge length L remain subject to the sizing closure of Sections 5.2 and 5.3.4, so its mass properties do not represent the budgeted values. Drawings of the frozen design will be issued in Appendix B.4.

Architecture: ballasted ring. Rather than adjusting inertia by scaling a solid disc, the wheel is built as a printed *ring plus spokes* whose inertia is then tuned by *ballast*: standard M6 steel bolt-and-nut sets fitted through round clearance holes drilled through the full ring thickness, distributed around the ring at a common pitch circle. Each station both removes plastic and adds steel at the largest available radius, so the net inertia gain per station is large, and the target is reached by choosing how many to install.

The bolt passing through the ring is also the retention feature: the nut threads onto it and clamps the ring, so nothing depends on a plastic pocket floor. Each station carries the centrifugal load $F = m_{\text{hw}}\,\omega_{\max}^2\,r_b$ of its own hardware, reacted by the bolt in tension and by bearing of the head and nut faces on the printed material , and a nyloc or threadlocked nut is specified so that vibration cannot back it off.

The geometry is defined by the quantities of Table 15, all of which follow from the envelope closure of Section 5.2 and are recorded there once fixed.

Table 15: Reaction-wheel geometry and ballast parameters. Values are the current design point of the iteration described below; they are provisional while the structural mass of Table 13 remains an allowance.

Symbol	Parameter	Value
D_w	Ring outer diameter (fixed by envelope)	120 mm
D_i	Ring inner diameter ($D_w - 2w_r$)	90 mm
w_r	Ring radial width	15 mm
t_r	Ring axial thickness (= hole depth)	5 mm
n_s, w_s	Spoke count and width	3, 10 mm
ρ_p	Printed-material density (PETG-CF)	1290 kg/m^3
d_h	Ballast hole diameter (M6 clearance)	6.4 mm
r_b	Ballast pitch circle radius	52.5 mm
m_{hw}	Hardware mass per station (bolt + nut)	9.0 g
m_b	*Net* mass added per station	8.79 g
ΔI_b	*Net* inertia added per station	2.428×10^{-5} kg m^2
N	Station count reaching $I_{w,\text{target}}$	15
m_w	Resulting wheel mass	172.5 g
I_w	Resulting wheel inertia	4.599×10^{-4} kg m^2
	as a fraction of $I_{w,\text{target}}$	111.7 %

The ballast is characterised *net* of the plastic each hole displaces: m_b and ΔI_b are the increments the installed bolt-and-nut adds over the full-depth cylinder of material it replaces, evaluated at the pitch circle radius. Both corrections are taken about the spin axis by the parallel-axis theorem, retaining each feature's own-axis term rather than treating it as a point mass. Sizing then reduces to solving

$$I_w(D_w, N) \;=\; I_{w,\text{ring}}(D_w) + N\,\Delta I_b \;\geq\; I_{w,\text{target}}$$

for the smallest admissible N at each candidate D_w, subject to the fit and mass checks below.

Sizing calculator and iteration procedure. The relations above are implemented as a parametric calculator that maps the geometry of Table 15 onto the mass and inertia of one wheel, sweeping candidate outer diameters and returning, for each, the station count required, the count the gap rule admits, the resulting mass and inertia, and the pass/fail state of every constraint. It is not an optimiser, and the design point it reports was not found by one.

The outer diameter is not among the quantities it is free to choose. D_w is *imposed* by the packaging envelope—three orthogonal wheel modules and the battery must fit within L—so the sweep is retained for what it reveals rather than for what it selects. Left free to minimise wheel mass it prefers a substantially larger ring, since I_{zz} scales as mR^2 and mass placed at a large radius buys inertia quadratically more cheaply; at 120 mm the constraint therefore costs about 85 g per wheel against that unconstrained optimum. This is the price of a wheel that fits, and a wheel that does not fit is not a candidate.

The procedure actually followed is the fixed-point iteration of Table 11, executed by hand. A cube mass is estimated and a mass budget assembled from it; the jump-up analysis converts that mass into an inertia requirement; the wheel is sized against the requirement subject to the internal-volume and hole-pattern constraints; and the resulting wheel mass is returned to the budget, which changes the mass that started the loop. The cycle repeats until it lands on a geometry that simultaneously fits the internal volume, admits a station count satisfying the inertia requirement, and reproduces the cube mass it assumed. Convergence is what closes the design: the reported point is one at which the assumed and the totalled mass agree, so step 6 of Table 11 returns approximately zero.

Selection among admissible diameters is made on minimum *wheel mass* rather than minimum diameter.

5.4 Electrical Design

The electrical architecture is presented here at the level of the four decisions that shape it: what the components are, how power is distributed, how the buses are routed and terminated, and how the whole is made safe to power on. The sheet-level schematics, the connector and pin tables, the harness drawings and the perfboard layout are collected in Appendix C, for the same reason the CAD detail is separated from the mechanical rationale — a connector re-pin should not require an edit here.

5.4.1 Components

The electronics and software will be built around a microcontroller communicating over a CAN bus with a motor driver that closes the current loop for the selected brushless motor. Wheel speed will be measured with a magnetic encoder on the motor shaft, and body tilt will be estimated from an onboard IMU. Braking will be actuated by a digital servo driven directly from the microcontroller, and a wireless module will provide a telemetry and debugging link. Power will be supplied by a LiPo battery, stepped down through a DC-DC converter for the logic-level electronics, with the motor driver powered directly from the battery bus.

The full parts list is catalogued in Appendix A (Table 20). The supplied set is an electronics-only BOM; the reaction wheels, the frame and all printed parts are the team's design responsibility, and their fabrication is the leading schedule risk.

Motor characterisation and operating envelope. Figure 8 shows measured torque and speed against phase current for the MN4006, taken from manufacturer bench data. The measurements were made with the motor driving propellers rather than reaction wheels, so the *load* is not representative of this application — an aerodynamic load rises with the square of speed, whereas a reaction wheel is an inertia with only bearing friction opposing it at constant speed. What does transfer is the motor's own electromechanical behaviour, since torque per ampere and back-EMF per unit speed are properties of the machine and not of what it is attached to. The curves are used here only in that sense: to bound the torque and current the actuator can be expected to deliver.

Three figures relevant to the design follow from the data. First, the torque–current relation in Figure 8(a) is close to linear ($r = 0.97$) with an incremental slope of 0.023 N m/A and a positive

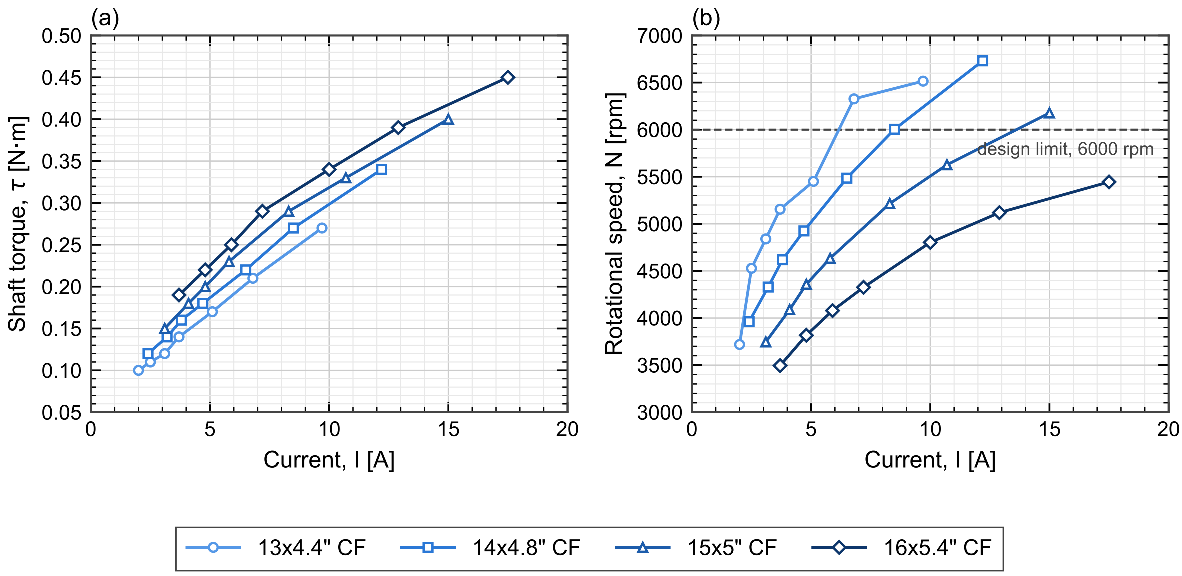


Figure 8: Measured MN4006 (a) shaft torque and (b) rotational speed against phase current, for one motor driving four different loads. Symbols are measured points at throttle settings of 50, 55, 60, 65, 75, 85 and 100 %; lines join successive points and are not fits. The dashed line in (b) is the 6000 rpm reaction-wheel design ceiling of Section 5.2. Loads shown are propellers, not reaction wheels: the curves characterise the *motor*, and the load-dependent spread between them does not carry over to this application. Manufacturer static bench data; the heaviest load reached the motor thermal limit at 100 % throttle.

torque-axis intercept of about 0.08 N m. The intercept is the expected signature of losses that do not reach the shaft — magnetising and iron losses plus bearing friction — so the useful torque per ampere is the slope, and a constant inferred by simply dividing torque by current at a single point would overstate it (that ratio runs to 0.031 N m/A through the origin). Either way the motor produces at most 0.45 N m, an order of magnitude below the 5 N m brake torque assumed in Section 4.3.1. The two are not alternatives: the motor stores angular momentum in the wheel over about a second, and the brake destroys it in tens of milliseconds, so the impulsive torque comes from the collision and not from the drive. This is why braking is a separate mechanism rather than a motor function. Second, the highest measured point reaches 0.45 N m at 17.5 A, above the 16 A peak rating in Table 20, and that run terminated on the motor's thermal limit — so the top of this envelope is a short-duration capability, not a continuous one. Third, speeds cross the 6000 rpm design ceiling of Section 5.2 only under the lighter loads, which is consistent with that ceiling being set by bus voltage and derating rather than by any inability of the motor to spin faster.

5.4.2 Power Architecture

The power architecture is shown in Figure 9. A single 6S LiPo pack (22.2 V nominal, 1550 mA h) feeds everything through one XT60 connector, behind a combined E-stop and fuse, with an anti-spark path to limit inrush into the driver bulk capacitance. The XT60 is fixed by the pack rather than chosen: it is rated at ≈ 60 A, which covers the three-axis spin-up transient but with a smaller margin than a 90 A housing would give, so the transient is a measured acceptance item at first power-on rather than an assumed one. The bus then splits into two rails. The *motor bus* runs at pack voltage (22–25 V across the discharge curve) directly into the three moteus drivers, with bulk capacitance rated to at least 50 V to absorb regenerative braking transients—which are not incidental here, since the jump-up manoeuvre of Section 4.3 dumps the full wheel kinetic energy back into the bus. The *logic rail* is derived by an LM2596 buck converter stepping 22 V down to 5 V for the Teensy, the ESP32 telemetry link, the three brake servos and the IMU. Keeping the two rails separate downstream of the E-stop means motor-current ripple does not reach the sensing and processing electronics.

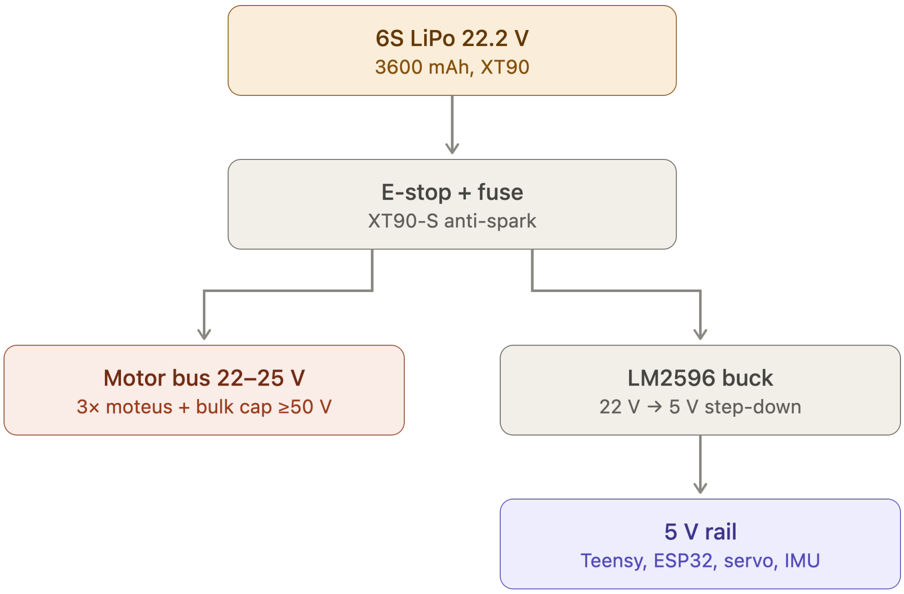


Figure 9: Power distribution tree. A single 6S pack feeds the motor bus directly and the 5 V logic rail through a buck converter, both behind a common E-stop and fuse.

5.4.3 Schematic and Interconnect Diagrams

The signal-level design is documented as a set of sheets rather than a single drawing, so that each can be checked against the constraint that governs it. Table 16 is the index; the sheets themselves are in Appendix C.1, together with the connector and pin-assignment tables. The design is currently captured as one flat schematic (Figure 12), from which these sheets are split as each is verified against its governing constraint; the index below is therefore the release plan as much as it is a table of contents.

5.4.4 Bus Topology, Termination and Signal Integrity

The CAN-FD bus is the one electrical subsystem whose physical routing is a correctness requirement, and it is therefore fixed before the board outline is drawn. At the design data rate the bus must be a linear chain with short stubs and termination at exactly the two physical ends; a star topology or a mid-chain termination will pass a bench continuity check and then fail intermittently under wheel vibration, which is the hardest class of fault to diagnose once the cube is closed. The design rules and their verification are listed in Table 17.

The encoder interfaces impose the complementary constraint. Each absolute encoder requires a short SPI run to its own driver and a tightly held air gap to its magnet, so the driver positions are dictated by the encoder positions, and those in turn by the motor axes. The physical layout order is therefore: motor axes fix the encoders, encoders fix the drivers, drivers fix the bus route, and only then is the board outline of Section 5.4.5 drawn around what remains.

Table 16: Schematic sheet index. Each sheet is checked against the governing constraint stated in its row before release.

Sheet	Content	Governing constraint to verify
1	Power entry and protection	E-stop, fuse rating, anti-spark inrush into the driver bulk capacitance
2	5 V logic rail	Converter rating vs. Table **??** worst case; back-feed diode; rail bulk capacitance voltage rating
3	CAN-FD bus	Linear topology, stub length, split termination at the two physical ends only (Sec. 5.4.4)
4	Encoder interfaces (3×)	SPI cable length and encoder air gap; driver placement follows from this, not the reverse
5	Flight computer and IMU	Sensor orientation and mounting location relative to the pivot (Sec. 4.4)
6	Telemetry link	Transmit-burst current and its return path off the logic rail
7	Brake servo drive	Stall current headroom on the logic rail; PWM routing away from the encoder SPI

Table 17: Signal-integrity design rules and how each is verified. These are checked before cube closure, since all are hard to access afterwards.

Interface	Rule	Verification
CAN-FD bus	Linear chain; short stubs; split termination at the two physical ends only	Differential-pair scope capture; sustained error-frame count at full rate with all nodes addressed
Encoder SPI	Short run to its own driver; air gap and chip centring per the assembly spec	Read-back with no error flags; re-checked after mechanical assembly
IMU	Mounted at or near the pivot; orientation per the body frame of Sec. 5.3.1	Static orientation check; noise floor with wheels spinning
Rail separation	Motor bus and logic rail separated downstream of the E-stop	Ripple measurement on the logic rail under motor load
Grounding	Single return topology; no motor return current through the sensing ground	Continuity and isolation check before first power-on

5.4.5 Board Layout and Harness

The electronics are built on a double-sided perfboard rather than a fabricated PCB. A PCB would have a long lead time, and a perfboard allows for iteration and modification throughout the process. Its consequence is that layout discipline — rail separation, decoupling at each device, deliberate routing of the bus pair — must be enforced by hand and checked explicitly, since no design-rule check will do it.

The board outline, mounting-hole pattern, keep-out heights and connector exit directions constitute the electronics-mount interface of Table 14 and are frozen once delivered to the mechanical design: later electrical changes must fit inside that outline rather than move it. The harness — battery lead, driver drops, bus chain and servo drive — is fabricated to the drawings in Appendix C.3 and is labelled and strain-relieved before installation, because every connector in the machine sits on a structure that vibrates by design.

5.4.6 Electrical Safety and Bring-Up Protocol

Two rules govern all electrical bring-up and are stated here because they are design constraints, not merely lab practice. First, every first-time power application is made on a current-limited bench supply, never on the battery; the battery is introduced only after the assembly has passed a full bench power-on. Second, the pack is never connected without the E-stop and fuse in circuit and the anti-spark path intact, since the driver bulk capacitance draws an inrush that will weld a bare connector.

The staged sequence — rails, then buses, then sensors, then open-loop motion, then closed-loop — is part of the validation plan . The isolation checks that precede first power-on (rail-to-rail separation, bus-to-chassis, and confirmation that no low-voltage capacitor sits on the pack bus) are recorded in Appendix C as a checklist, because they are performed on every re-assembly and not only once.

5.5 System Integration

5.5.1 Onboard Software Pipeline

The subsystems of Figure 5 are coordinated by a single deterministic task on the Teensy, whose structure is given in Figure 10. Each 1 ms cycle executes the same fixed sequence: sensor ingest reads the BMI270 over SPI and the moteus replies over CAN; the MEKF propagates attitude on the gyro and corrects it against the accelerometer while estimating gyro bias; the state vector $x = [g_B, \omega, h]$ is assembled from the estimator output and the wheel speeds; the LQR law of Section 4.4 evaluates $u = -Kx$ with a penalty on stored momentum h to bleed off wheel saturation; and the resulting commands leave as CAN torque requests to the drivers and a PWM signal to the brake servos. Telemetry is tapped off the assembled state to SD and Wi-Fi outside the control path, so logging cannot perturb loop timing.

The fixed ordering matters: with no dynamic allocation and no blocking calls, the worst-case cycle time is bounded by construction rather than measured after the fact, which is what makes the 1 kHz sample period of the discrete LQR design a design guarantee rather than an average.

5.5.2 Control Software Architecture

Section 4 derives the control law; this subsection states how that law is realised as software, which is a separate set of decisions. The implementation detail — the mode state machine, the estimator and controller in executable pseudocode, the fixed gains and tuning parameters, the timing budget and the fault responses — will be collected separately, so that a re-tuned gain does not require an edit to the design sections.

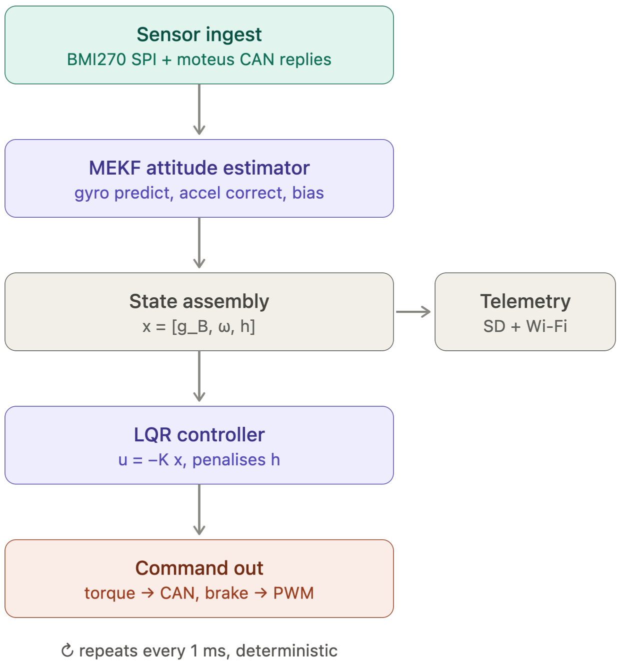


Figure 10: Teensy 4.1 control-loop software pipeline. The estimator and controller stages implement the design of Section 4.4; telemetry is a side-branch off the control path.

The software is organised in the four layers of Table 18. The division is not stylistic: each layer has a different failure mode and a different verification method, and the interfaces between them are what allow the controller validated in simulation to be the same code that runs on the cube.

Table 18: Control software layers. Each is verified by a different method, which is why the boundaries are held.

Layer	Responsibility	Verified by
Hardware abstraction	Sensor and driver I/O, timing, bus transactions	Bench read-back; loop-timing measurement
Estimation	Attitude and rate estimate from IMU and encoders (Sec. 4.4)	Static and dynamic bench tests against a known reference
Control	State feedback, momentum management, saturation handling	Simulation on the identified model, then hardware
Supervision	Mode transitions, arming, watchdog and fault response	Fault-injection tests on the rig

Operating modes. The supervisory layer sequences the modes of Table 19, with hysteresis on every threshold to prevent chattering at a boundary, and with a disarm transition available from every mode. The modes are specified as a state machine because the critical transitions are the descending ones: a fall detected during corner balance must reach a safe state with the wheels commanded down, not merely switch to a different gain.

Real-time guarantees. The loop-rate claim of Section 4.4 is a design guarantee only if the worst-case cycle is bounded by construction, which is why the pipeline of Figure 10 uses a fixed stage ordering with no dynamic allocation and no blocking calls, and why telemetry is a side-branch rather than a stage.

Table 19: Operating modes and transitions. Every mode has a disarm exit.

Mode	Behaviour	Exit condition
Idle / disarmed	Drivers de-energised; sensors and telemetry live	Explicit arm command
Wheel spin-up	Open-loop speed ramp to the jump-up momentum target	Target wheel speed reached
Jump-up	Brake actuation and momentum transfer (Sec. 4.3)	Tilt inside the balancing capture band
Edge balance	Single-axis state feedback (Sec. 4.4)	Corner-transition command, or fall detected
Corner balance	Three-axis state feedback (Sec. 4.5)	Slew command, or fall detected
Slew / rotation	Reference tracking between equilibria	Reference reached; revert to balance
Fault / safe	Wheels commanded down, drivers disarmed	Manual reset

Fault handling. The failure cases the software must survive are sensor dropout, bus error, wheel saturation and loss of the balancing equilibrium. The first three have defined responses that keep the machine controlled; the fourth is a controlled shutdown rather than a recovery attempt, since a cube that has left its capture region cannot be recovered by the linear law and continued actuation only adds energy to the fall.

6 Future Work

6.1 NMPC & Warm-Start Control (Future Implementation)

6.2 Autonomous Stand-up & Disturbance Rejection

6.3 AI-Driven Control & Dynamic Coefficient Tuning

6.3.1 Motivation: Limitations of a Fixed Gain

The balancing controller of Section 4.4 computes a single gain K_d offline, from an (A, B) pair built from the parameters of Tables 5 and 10. That gain is optimal for exactly one plant: the one those numbers describe. It is retained here as the baseline precisely because it is simple and verifiable, but its optimality is conditional in three ways that the hardware of this project is expected to violate.

- **Parametric drift.** C_b and C_w are not obtained from CAD and are carried as estimates until the identification tests of Section **??** (Table 6). Bearing friction, wiring drag and motor cogging also change with temperature, wear and reassembly, so the identified values are a snapshot rather than a constant. The inertia tensor I and the centre-of-mass offset r_{cm} are subject to the same objection at the few-percent level, and r_{cm} in particular shifts whenever the pack or harness is disturbed.
- **Payload and configuration changes.** Any added mass changes m, r_{cm} and I simultaneously, and therefore changes A. A gain tuned for one configuration is not merely suboptimal for another; near a fully unstable equilibrium it may be destabilising.
- **Contact and surface variation.** The pivot is modelled as an ideal frictionless point or line. A real corner resting on a real surface slips, deforms and dissipates energy in a way that is neither in the model nor constant across surfaces, and this is the dominant unmodelled effect for the walking objective of Section 1.3.

The standard remedy within the existing design is gain scheduling on $\hat{\theta}_b$ (Section 4.4), but scheduling interpolates between gains that were themselves computed from the same assumed parameters. It addresses the weakening of the small-angle linearisation; it does not address a plant that differs from the model. The proposal here is to keep the verified feedback law and adapt its coefficients online.

6.3.2 Proposed Hybrid Architecture

The intended structure is deliberately conservative: a classical inner loop whose stability properties are understood, wrapped by a slow outer loop that adjusts the inner loop's coefficients. Writing ϕ for the tunable coefficient vector,

$$u[k] = -K_d(\phi[k])\,\hat{x}[k], \qquad \phi[k+1] = \phi[k] + \Delta\phi\big(\hat{x}[k],\, \hat{x}[k-1], \dots\big), \tag{17}$$

where $\hat{x}$ is the estimated state of Section 4.4 and $\Delta\phi$ is produced by the tuner. Two properties of (17) matter more than the choice of learning algorithm. First, the inner loop remains a linear state feedback at every instant, so the controller that runs on the hardware is the one already implemented and tested. Second, ϕ is updated on a timescale far slower than the control period Δt, which keeps the frozen-gain analysis approximately valid and prevents the tuner from acting as an unmodelled fast feedback path. Three candidate tuners are of interest, and they are not equivalent in cost:

- **Bayesian optimisation** over a low-dimensional parameterization of Q and R, minimising a measured cost from short balancing trials [**shahriari2016bayesopt**]. This is the cheapest option and the one with a direct precedent on this class of hardware: automatic LQR tuning has been demonstrated on an inverted pendulum with on the order of tens of experiments [**marco2016automaticlqr**], and a self-tuning LQR has been run on the Cubli's own ancestor platform at ETH [**trimpe2014learningcubli**]. It tunes between trials, not within them.

- **Adaptive dynamic programming**, which solves the LQR problem online from measured data rather than from (A, B), converging toward the optimal gain for the true plant without an explicit model [**lewis2012adp**]. This is the option with the strongest theoretical guarantees and the strictest excitation requirements.
- **Reinforcement learning**, in which a policy trained in simulation outputs $\Delta\phi$ continuously from the state-error history. This is the most capable and the least verifiable, and is treated below as the exploratory branch rather than the default.

Deployment reservations. The literature supporting learned control is overwhelmingly validated in simulation or on well-instrumented hardware, and the failure mode of an online tuner on an open-loop-unstable plant is a fall, not a degraded score. Any deployment on this cube should therefore be gated behind: a projection of ϕ onto a set for which the frozen-gain closed loop is verified stable across the parameter uncertainty; a fallback to the fixed K_d of Section 4.4 on any estimator fault or divergence; and bench trials with the cube tethered. This is future work in the literal sense — none of it is a committed deliverable of the present design, and the baseline controller is not contingent on any of it.

6.3.3 Module 1: Dynamic Gain Tuning

The tuner does not emit motor currents. It emits the coefficients from which K_d is computed, which bounds what a bad update can do. Two parameterizations are worth comparing:

- **Weight-space tuning.** The policy outputs the diagonal entries of the cost matrices, $\phi = (\log q_1, \ldots, \log q_9,\ \log r_1, \log r_2, \log r_3)$ with $Q = \operatorname{diag}(q_i) \succ 0$ and $R = \operatorname{diag}(r_i) \succ 0$, and K_d follows from the discrete algebraic Riccati equation. The logarithmic parameterization enforces positive definiteness by construction, so every ϕ the policy can emit corresponds to a well-posed LQR problem and, for a stabilisable (A_d, B_d), to a stabilising gain for the *modelled* plant. The cost is that a Riccati solve must run onboard whenever ϕ changes, which is what confines updates to a slow outer loop.
- **Gain-space tuning.** The policy perturbs the entries of K_d directly, or equivalently the K_p, K_d pair of a PD form, avoiding the Riccati solve at the price of losing the structural guarantee above. This is the practical choice if the outer loop must run fast, and it makes the projection step of the previous paragraph mandatory rather than merely prudent.

For the 3D corner-balancing case the state is $x = (\theta, \omega_b, \omega_w) \in \mathbb{R}^9$ and K_d is 3×9 (Section 4.5), so tuning the 54 free entries directly is not sensible on the available data. Weight-space tuning reduces the problem to 12 coefficients, and symmetry of the three wheel axes reduces it further if the wheels are treated as identical. A reward or cost of the form

$$c[k] = \hat{x}^{\mathrm{T}}[k]\, W_x\, \hat{x}[k] + \rho\, \|u[k]\|^2 + \sigma\, \|\omega_w[k]\|^2 \tag{18}$$

adds an explicit penalty on stored wheel momentum, which the nominal LQR cost $J(u)$ of Section 4.4 does not distinguish from any other state deviation. This term is the one directly relevant to the saturation problem raised in Section 1.2: it expresses a preference for holding attitude with the wheels near zero speed, leaving momentum headroom for disturbance rejection.

6.3.4 Module 2: Sim-to-Real Transfer via Domain Randomization

A policy trained against a single nominal parameter set learns to exploit that parameter set. Domain randomization instead samples the plant parameters from a distribution at the start of each training episode [**tobin2017domainrand**], so that the policy is optimised against a family of plants rather than one, and the transfer method has a substantial record on physical legged hardware [**hwangbo2019anymal**]. For this cube the randomized quantities follow directly from the identification gaps already documented:

- **Damping:** C_b, C_w over a wide interval, since these are estimated rather than measured until Section **??** closes.
- **Inertia and mass distribution:** I, m and r_{cm} perturbed by a few percent, covering print-density variation, fastener placement and harness routing.
- **Actuation:** K_m within motor tolerance, plus a current saturation limit and a commanded-torque delay, so the policy cannot assume an ideal actuator.
- **Contact:** pivot friction and restitution, and a small random ground slope — the term standing in for the surface variation that motivates this module.
- **Sensing:** gyro bias walk, accelerometer noise and an injected vibration component, matching the estimator degradation identified in Section 1.1 as the practical limit on balancing.

The sensing term requires particular care. The accelerometer corruption on this platform is generated by the controller's own wheel activity, so it is correlated with the policy's actions rather than independent of them. A randomization scheme that injects only white sensor noise will understate the difficulty, and a policy that transfers under it should not be assumed to transfer on hardware. Whether the randomized distribution covers the real cube is an empirical question, settled by bench testing; the principal failure mode of this module is a distribution selected for convenience rather than representativeness.

6.3.5 Module 3: Jump-Up Energy Optimisation

The jump-up analysis of Section 4.3.1 treats the manoeuvre as a spin-up to $\omega_{\max}$ followed by an abrupt stop, with all deviation from the ideal absorbed into a single brake-amplification factor $\beta = \tau_b/(\tau_b - \tau_g)$. That model is adequate for sizing and is deliberately conservative, but it is a poor description of what the mechanism does: the servo-driven barrier has finite travel, the collision is not instantaneous, and the resulting torque profile is neither a step nor known in advance.

The proposal is to treat the torque command over the manoeuvre window as a profile to be optimised rather than a constant. Parameterizing the pre-brake and post-impact torque commands by a small vector ψ — for instance the timing of brake release relative to the wheel-speed trajectory, and the shape of the recovery torque applied by the remaining wheels immediately after impact — the objective is

$$\min_{\psi} \; \mathbb{E}\Big[\|\theta(t_f) - \theta^\star\|^2 + \mu \, \|\omega_b(t_f)\|^2 \Big] \quad \text{subject to} \quad \omega_w \le \omega_{\max}, \quad |u_i| \le u_{\max}, \tag{19}$$

where t_f is the instant the controller hands over to the balancing law at $|\hat{\theta}_b| < \theta_{\text{th}}$. Minimising residual body rate at handover is the quantity of interest, because a jump that reaches the equilibrium with excess rate demands recovery authority the balancing controller may not have. Because each evaluation of (19) is a physical jump, the sample budget is small and the appropriate method is Bayesian optimisation rather than a policy-gradient method, with the initial design taken from the analytical h_w of Equation (4).

Two constraints on this module must be stated. First, it presupposes the measurement described in Section 4.3.1: until the spin-down test of Section **??** reports both the impulse-equivalent mean $\bar{\tau}_b = h_w/\Delta t$ and the peak torque, there is no validated model to optimise against and no way to distinguish an improved profile from a mis-measured one. Second, the structural caveat carries over unchanged — optimisation must not be permitted to search toward shorter stopping times, since τ_b is simultaneously a load on the barrier, bolt, bracket and spokes, and the sizing value was chosen for the dynamics and not for that load path. The search space must be bounded by the structural case, not by the dynamic one.

6.3.6 Relevance Beyond the Bench

Two application claims follow, and both are narrower than the general enthusiasm for learned control would suggest.

For **CubeSat attitude control**, the relevant fact is that a spacecraft's inertia tensor is known imprecisely at integration and changes in flight — propellant depletion, deployable appendages, thermal distortion — while reaction-wheel friction rises with bearing age. These are the orbital analogues of the parametric drift of Section 4.4, and they are conventionally handled by conservative fixed gains with margin, at a cost in pointing performance. A tuner that adapts Q and R against measured performance addresses exactly this gap. The qualification barrier for flying learned control is high and unaddressed here; the architecture of (17) is chosen partly because a bounded outer loop over a classical inner loop is the form most likely to survive that scrutiny.

For **low-gravity surface mobility**, the case is stronger, because the plant is genuinely unknown before arrival. Regolith mechanical properties on a small body are not characterised to the precision a hopping controller would want, and the flight-proven hoppers cited in Section 1.2 [5, 6] move ballistically rather than under closed-loop control for this reason. The contact randomization of Module 2 and the profile optimisation of Module 3 map onto that problem directly: a hop whose landing attitude is controlled, on a surface whose properties were sampled from a distribution rather than assumed, is the capability that separates directional walking from a ballistic tumble [7]. The cube tests this under 1 g, where the destabilising torque is larger and never relents — a harder case in that one respect, as argued in Section 1.2, though it does not reproduce the low-gravity contact dynamics that would decide the real manoeuvre.

6.4 Single-Wheel Variant / Extensions

References

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A Bill of Materials

The components supplied for this project constitute an *electronics* bill of materials: the actuation chain, wheel and attitude sensing, on-board compute, communications and power distribution are specified, but no structural or mechanical hardware is. The reaction wheels, the frame, the motor mounts, the fasteners and every printed bracket are the team's design responsibility, and are treated as such in Section 5. Table 20 lists the supplied items grouped by subsystem. The actuation figures derived from these parts — the torque constant, the achievable torque and wheel speed, and the inertia that sizes the wheel — are developed where they are used, in Sections 4.3.1 and 5.3.4.

The architecture the BOM implies is a two-level control stack: each mjbots moteus-n1 closes a high-rate field-oriented current loop on its motor, using the MA600 encoder for rotor angle, while the Teensy 4.1 runs the outer attitude-estimation and state-feedback loop over CAN-FD — the same inner-tachometer / outer-attitude layering used in spacecraft reaction-wheel assemblies. A mechanical wheel brake, following the original ETH Cubli [2], provides the near-impulsive momentum transfer required for the jump-up.

Procurement is the single largest schedule risk. The long-lead items are not electronic but mechanical: the frame stock and the printed structure gate the entire build, and with three actuation sets and no spare, the loss of a motor or driver in service would force the reduced-actuation one-wheel variant [8]. These items are ordered first. Consumables and bench equipment not covered by the supplied set — wire, connectors, fasteners, filament, the current-limited bench supply and the battery-safety kit — are tracked separately in the build documentation.

Table 20: Supplied bill of materials, grouped by subsystem.

Ref	Item	Qty[a]	Key specifications	Function
Actuation				
1	T-Motor MN4006 KV380	3	18N24P outrunner; 380 rpm/V; 68 g; ⌀44.35 mm × 21 mm; 194 mΩ; peak 16 A, 380 W	Reaction-wheel torque source
2	mjbots moteus-n1	3	10–54 V; 100 A peak / 9 A cont.; 15–30 kHz FOC; 5 Mbit/s CAN-FD; 14.6 g	FOC current/torque driver (inner loop)
11	TowerPro MG92B servo	3	metal-gear micro servo; ≈0.29 N m (≈3 kg cm) at 6 V; 50 Hz PWM	Wheel brake actuator (jump-up)
Sensing				
3	mjbots MA600 breakout + ring magnet	3	TMR absolute angle; SPI 6 MHz, 3.3 V; 16-bit (65,536 counts/rev); air-gap ≤ 0.5 mm	Wheel angle: commutation + state feedback
10	SparkFun BMI270 6-DoF IMU	1	3-axis accel + 3-axis gyro (no magnetometer); I^2C 400 kHz (Qwiic); ODR up to ≈ 1 kHz	Body attitude reference (tilt + rates)
Compute and Communications				
4	Teensy 4.1	1	600 MHz Cortex-M7 + FPU; native CAN-FD (CAN3); 5 V supply	Flight computer: estimator + control law
5	CAN-FD adapter (Teensy 4.1)	1	CAN-FD transceiver, logic-level → differential; 5 Mbit/s	Bus transceiver for the MCU

Continued on next page

Table 20 (continued)

Ref	Item	Qty[a]	Key specifications	Function
6	mjbots mjcanfd-usb-1x	1	USB $\leftrightarrow$ CAN-FD, 5 Mbit/s	Bench interface (calibration, tuning)
7	JST-PH3 CAN cables	$\geq$4	2 A AC/DC; 3-conductor daisy-chain	CAN bus interconnect
8	Seeed XIAO ESP32-C6 + antenna	1	Wi-Fi; 3.3 V UART to MCU; TX burst $\approx$300 mA	Wi-Fi telemetry / command link
9	60.4 Ω 0.5 W resistor	4	Split termination, $2 \times 60.4\,\Omega \approx 120\,\Omega$ bus impedance	CAN bus termination
17	4.7 nF capacitor	2	common-mode filter, bus midpoint to ground	CAN split-termination filter
Power				
12	Tattu R-Line V5.0 6S LiPo (XT60)	1	22.2 V; 1550 mA h; 150C; $\approx$35 W h; $\approx$254 g	Primary energy store
13	LM2596 buck module	1	step-down to 5.0 V; rated 2–3 A (derates without heatsink)	5 V logic rail
14	XT60 connector pair	1–2	rated $\approx$60 A	Battery power connector
15	100 µF 16 V electrolytic	2–4	16 V part — 5 V rail only	5 V-rail bulk reservoir
19	1N5819 Schottky diode	1–2	1 A, 40 V	5 V-rail back-feed protection
Integration				
16	100 nF 50 V capacitor	6–10	local decoupling at each IC	Per-IC supply decoupling
18	Double-sided perfboard	1	15 cm × 9 cm, double-sided	Power / signal distribution board

[a] Quantity per assembled cube; motors, drivers, encoders and brake servos are 3 each — one per wheel axis.

B CAD and Construction Detail

This appendix holds the mechanical detail deliberately kept out of Section 5.3: the model tree, the drawing set, the fabrication settings and the assembly sequence. The rationale for the architecture — why the frame is split, what fixes the envelope, how the wheel is tuned — stays in Section 5.3 and is not repeated here. Anything in this appendix may change without the design intent changing; anything in Section 5.3 may not.

B.1 Design Parameter Table

Table 21 is the single source of the driving dimensions referenced by every part in the model, per convention 1 of Section 5.3.1. It is the CAD-side mirror of the envelope closure of Section 5.2 and is updated on each iteration.

Table 21: Driving CAD parameters. Every dependent feature references these; no part carries a duplicated hard-coded value.

Symbol	Parameter	Value
L	Cube edge length (outer)	15 cm
t_w	Frame wall thickness	5 mm
D_w	Wheel outer diameter (Sec. 5.3.4)	120 mm
t_r	Wheel axial thickness (Sec. 5.3.4)	5 mm
d_m	Motor axis offset from cube centre	TBD
h_m	Motor mounting face height on its axis	TBD
c_w	Wheel-to-subframe axial clearance	TBD
c_o	Clearance between orthogonal wheel modules	TBD
$V_{\min}$	Minimum internal clear volume	TBD
r_c	Corner contact feature radius	TBD

B.2 Model Tree and Part List

Table 22 lists the modelled parts by assembly, with their fabrication route and current status. It is the mechanical counterpart to the electronics bill of materials of Appendix A: the parts here are the team's design responsibility and are not supplied.

B.3 1-DoF Prototype Construction

The general-assembly drawing of the 1-DoF rig is given in Figure 11. This is the final released design of the single-axis prototype: the sheet carries the orthographic set — front and side elevations of the rig on its base, and a plan and edge view of the reaction-wheel assembly — from which the plate geometry, the pivot bearing seat, the motor bolt circle and the hard-stop placement are taken. The rig as built from this drawing is shown in Figure 6.

Table 22: CAD model tree and part list. Quantities are per assembled cube unless noted; the 1-DoF rig parts are listed separately.

Ref	Part	Qty	Fabrication route	Status
Inner structure				
M1	Motor subframe	TBD	TBD	TBD
M2	Motor mount (×axis)	3	TBD	TBD
M3	Battery cradle / retention	1	TBD	TBD
M4	Electronics tray	1	TBD	TBD
M5	Brake servo bracket	3	TBD	TBD
Reaction wheel assembly				
W1	Ballasted ring (Sec. 5.3.4)	3	TBD	TBD
W2	Wheel hub / rotor adapter	3	TBD	TBD
W3	Ballast bolt and nut set (M6, 3×15)	45	Standard hardware	TBD
W4	Brake strike feature	3	TBD	TBD
Outer frame				
F1	Frame member / edge rail	TBD	TBD	TBD
F2	Corner contact feature	8	TBD	TBD
F3	Face panel / wheel guard	TBD	TBD	TBD
1-DoF rig (Sec. 5.3.2)				
R1	Face plate	1	TBD	TBD
R2	Pivot bearing mount	1	TBD	TBD
R3	Rig motor mount	1	TBD	TBD
R4	Hard stops / containment	TBD	TBD	TBD

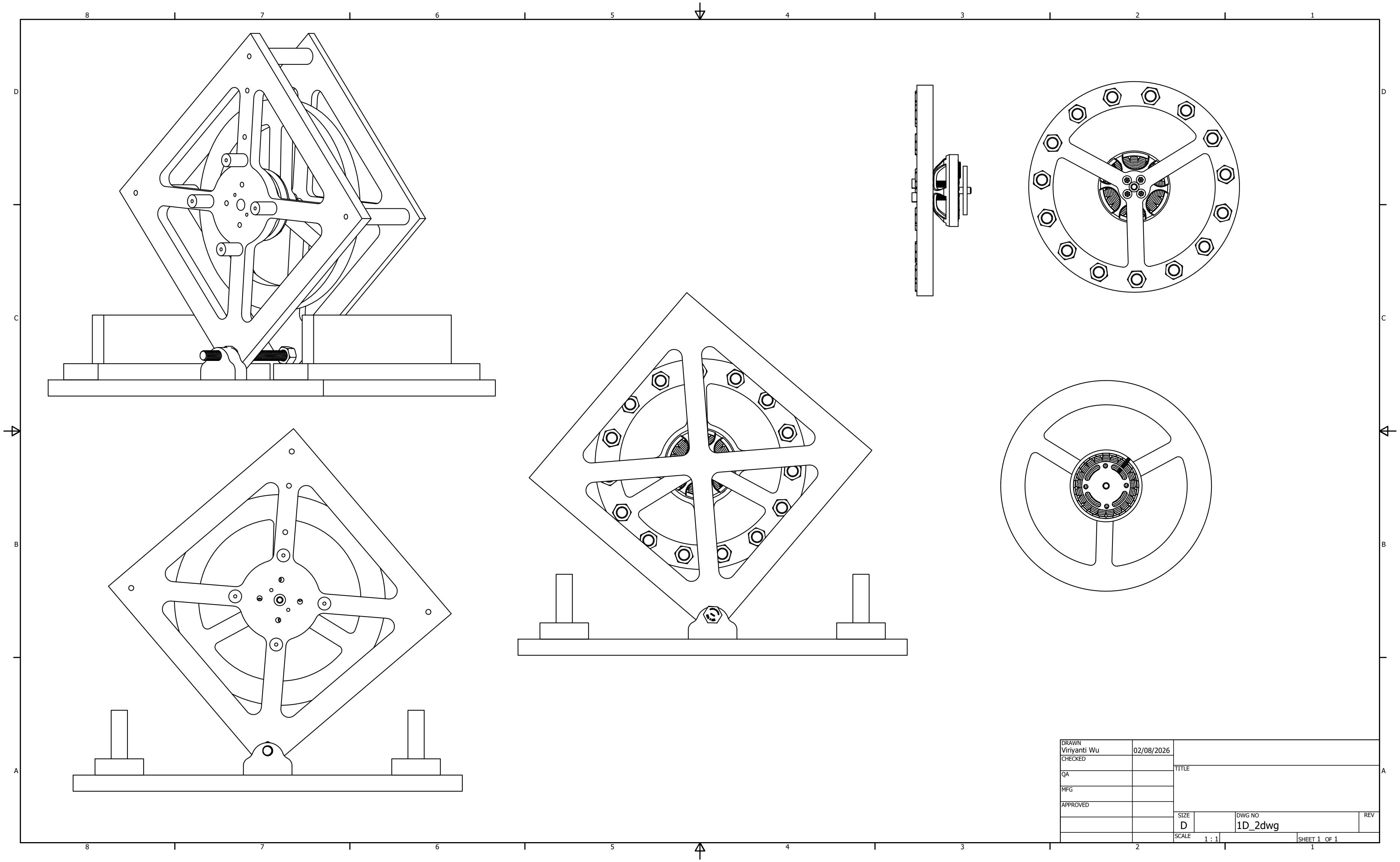

Figure 11: General-assembly drawing of the 1-DoF prototype rig — the final released design of the single-axis article — sheet 1 of 1 at 1:1. Upper left and lower left: front elevation of the rig mounted on its base, showing the diagonal plate orientation, the corner pivot and the two hard stops. Centre: side elevation through the wheel axis. Right: plan and edge views of the reaction-wheel and motor assembly, giving the spoke pattern and the ballast bolt circle used for the balancing adjustment of Section 5.3.4.

B.4 3D Cube Construction

The drawing set for the three-axis cube is not yet released. The frame has been realised so far as the concept prototype of Figure 7, which fixes the two-part architecture and the internal packaging but not the final dimensions: the edge length L and the wheel geometry close only once the sizing iteration of Section 5.3.4 converges on a confirmed motor. The drawings to be issued here on that closure are the inner-structure general assembly, the three motor modules on their orthogonal axes, the outer frame with its contact features, and a section view through two orthogonal wheel modules showing the clearances c_w and c_o of Table 21.

B.5 Fabrication and Assembly

B.5.1 Print and Manufacturing Settings

The printed parts are structural, not cosmetic: the wheel ring carries centrifugal load at the ballast holes and the motor mounts carry the brake impulse, which is the design load case identified in Section 1.1. Settings are therefore recorded per part in Table 23 and treated as part of the design, since a part reprinted at a different orientation or infill is not the part that was analysed.

Table 23: Fabrication settings for the structural printed parts. Layer orientation is specified relative to the principal load direction.

Ref	Material	Infill / walls	Orientation and rationale
W1	TBD	TBD	TBD
W2	TBD	TBD	TBD
M1	TBD	TBD	TBD
M2	TBD	TBD	TBD
F1	TBD	TBD	TBD
F2	TBD	TBD	TBD

B.5.2 Assembly Sequence

The sequence of Table 24 is ordered by access: each step is one whose verification becomes impossible or destructive once a later step is complete. The encoder air gap in particular is re-checked after mechanical assembly, since it can shift when the structure is bolted up (Table 17).

B.6 Mass Properties Verification

The CAD roll-up is an estimate until the assembly is weighed. Step 8 above closes the loop of Section 5.2: the measured mass is compared against the modelled value, and any discrepancy is carried back into $I_{w,\text{target}}$ and absorbed, where possible, by the ballast trim of Section 5.3.4 rather than by a re-design.

Table 24: Assembly sequence. Ordering is set by access: each check must be completed before the step that encloses it.

Step	Operation	Check before proceeding
1	Wheel assembly: ring, hub, ballast bolts to the selected count	Balance check; bolt retention per Sec. 5.3.4
2	Motor to mount, encoder magnet to rotor	Encoder air gap and centring to the assembly spec
3	Motor modules to inner structure, three axes	Axis orthogonality; wheel clearance c_w, c_o
4	Battery and electronics tray into inner structure	Pack centring; board keep-out clearances
5	Harness install and dressing	Continuity and isolation (Sec. 5.4.6)
6	Inner structure into outer frame	Encoder air gaps re-checked after bolt-up
7	Contact features and guards	Contact geometry; wheel guard clearance
8	Mass properties measurement	Measured mass against Table 13

Table 25: Modelled versus measured mass properties. A discrepancy here is absorbed by ballast trim where the margin allows.

Quantity	CAD value	Measured	Discrepancy
Total cube mass M	TBD	TBD	TBD
Wheel mass m_w (each)	TBD	TBD	TBD
Wheel inertia I_w (each)	TBD	TBD	TBD
CoM offset from cube centre	TBD	TBD	TBD

C Electrical System Detail

This appendix holds the electrical detail kept out of Section 5.4: the schematic sheets indexed in Table 16, the connector and pin assignments, the harness drawings, the board layout and the bring-up checklists. The architectural decisions — two rails behind one E-stop, linear bus topology, perfboard rather than fabricated board — remain in Section 5.4. The part specifications are in Appendix A and are not repeated.

C.1 Schematic Sheets

The system is captured on one flat KiCad sheet, reproduced at page size in Figure 12 and listed by functional block in Table 16. That capture is the working drawing the build is wired from. The blocks it carries are: power entry and protection (pack connector, anti-spark inrush path, combined E-stop and fuse, split to the motor bus and the logic rail); the 5 V logic rail off an LM2596 converter with back-feed protection and per-device decoupling; the CAN-FD bus as a linear chain across the three moteus drivers and the flight computer, terminated at the two physical ends only; the per-axis encoder SPI interfaces; the Teensy 4.1 flight computer with the BMI270 IMU; the ESP32 telemetry link; and the three brake servo drives.

Note: for the provided N35 1/4" $\times$ 1/8" diametric disk magnet, the optimal distance in order to achieve the 20 mT – 80 mT nominal magnetic field for the MA600 encoder is 3.3 mm.

The per-block schematic sheets, split out of this capture as a release step rather than a redraw, are to be added in the final draft.

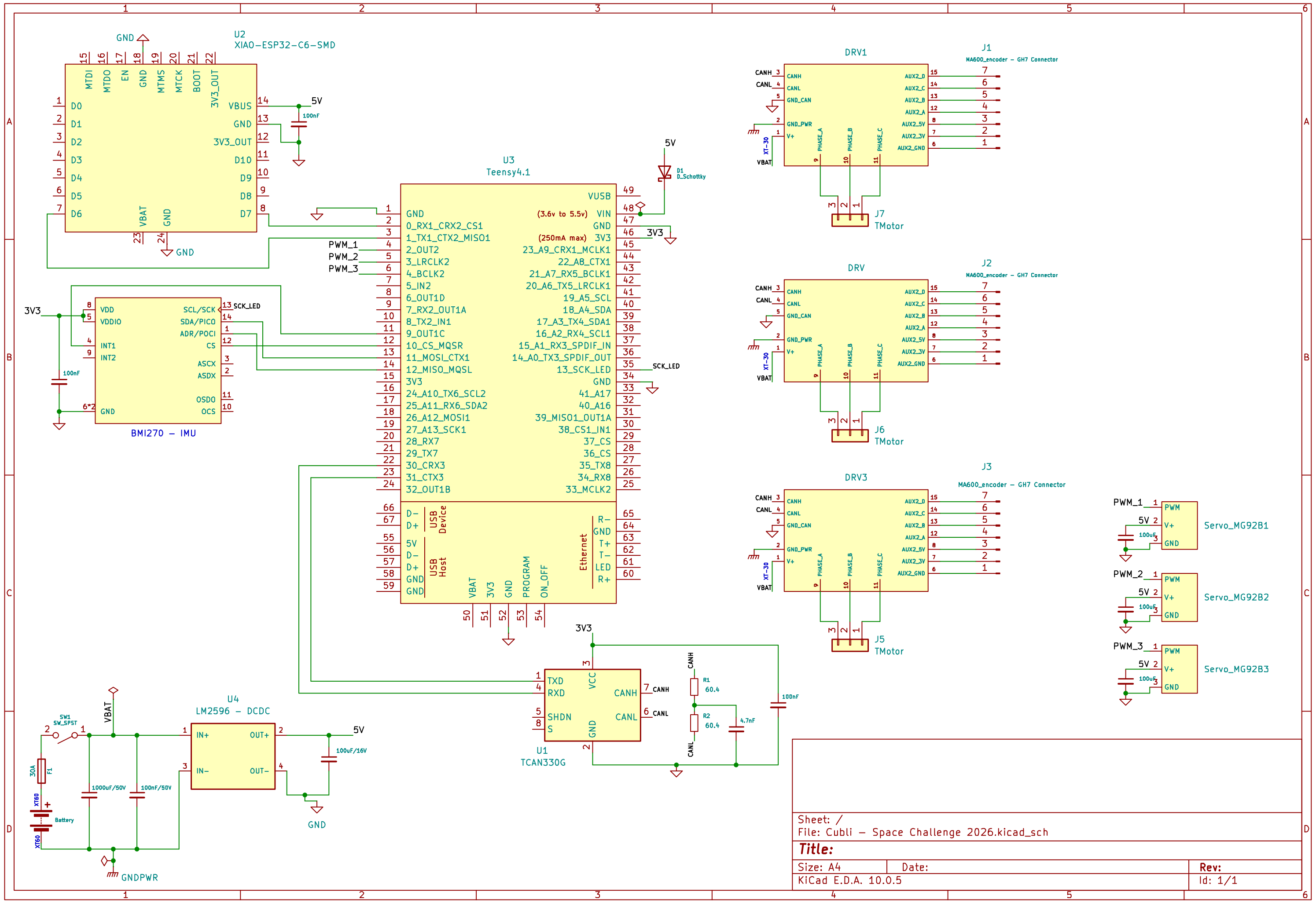

Figure 12: Full electrical schematic as captured, reproduced at page size. Covers the flight computer (Teensy 4.1) and its ESP32 telemetry module, the BMI270 IMU, the three moteus driver and motor connectors, the CAN-FD transceiver and termination, the LM2596 5 V converter from the pack, and the three brake servos. The per-block sheets are drawn from this capture as it is split for release.

C.2 Connector and Pin Assignments

Table 26 is the connector schedule. It exists so that a harness can be built and re-built from this document alone, and so that a mis-mate is a documented impossibility rather than a discovered one: no two connectors carrying different voltages share a housing type on this machine.

Table 26: Connector schedule. Keying and housing types are chosen so that two connectors carrying different voltages cannot be mated.

Ref	From	To	Housing / keying	Signals
J1	Battery pack	Protection / E-stop	TBD	Pack +, pack −
J2	Protection	Distribution board	TBD	Motor bus
J3	Distribution board	Driver, axis 1	TBD	Motor bus
J4	Distribution board	Driver, axis 2	TBD	Motor bus
J5	Distribution board	Driver, axis 3	TBD	Motor bus
J6	Board	CAN chain, end A	TBD	CAN-H, CAN-L, GND
J7	CAN chain, end B	Termination network	TBD	CAN-H, CAN-L, GND
J8	Driver, axis i	Encoder, axis i	TBD	SPI, 3.3 V, GND
J9	Board	IMU	TBD	Serial bus, supply, GND
J10	Board	Telemetry module	TBD	UART, supply, GND
J11	Board	Brake servo, axis i	TBD	PWM, 5 V, GND

C.3 Harness Drawings and Wire Schedule

Table 27 is the wire schedule. Conductor gauge on the motor bus is set by the spin-up current rather than the balancing current, since spin-up is the sustained high-current phase; the signal runs are set by length limits rather than by current.

Table 27: Wire schedule. Motor-bus gauge is sized on spin-up current; signal runs are length-limited (Table 17).

Ref	Run	Gauge	Length	Sizing basis
H1	Pack to protection	TBD	TBD	Peak bus current
H2	Protection to board	TBD	TBD	Peak bus current
H3	Board to driver drops (3)	TBD	TBD	Per-axis spin-up current
H4	Motor phase leads (3)	TBD	TBD	Motor phase current
H5	CAN chain segments	TBD	TBD	Topology, not current
H6	Encoder SPI runs (3)	TBD	TBD	Length limit governs
H7	Servo drives (3)	TBD	TBD	Stall current

C.4 Board Layout

The floorplan is constrained before it is aesthetic: the driver positions follow from the encoder cable limit, the bus route follows from the driver positions, and the board is placed around what remains (Section 5.4.4). Rail separation is maintained physically on the board, not only schematically, so that motor-current return does not share copper with the sensing ground.

C.5 Bring-Up and Isolation Checklists

These checks are performed on every re-assembly, not only on first build, since the encoder gaps and connector seating are exactly what mechanical work disturbs.

Before any power is applied.

1. Isolation: motor bus to chassis, logic rail to motor bus, each rail to ground.
2. Capacitor voltage ratings confirmed by position — no low-voltage part on the pack bus.
3. Polarity of every fabricated connector verified against Table 26.
4. Protection element in circuit: E-stop functional, fuse fitted and rated, anti-spark path intact.
5. Encoder air gaps re-measured after mechanical assembly.

First power-on, current-limited bench supply.

1. Current limit verified against a known load before connecting the assembly.
2. Logic rail voltage and ripple under load; converter thermal check.
3. All drivers enumerate on the bus; encoders read back with no error flags; IMU responds.
4. Bus integrity at the full data rate: sustained soak with all nodes addressed, error-frame count logged.
5. Open-loop wheel motion, wheels contained, one axis at a time.

First battery power-on. Performed only after the bench sequence passes in full, with the pack protection verified, cell monitoring active and the assembly in a fire-safe area. Inrush is measured against the bulk capacitance on connection.

C.6 As-Built Measurements

Table 28 records the measured electrical figures that replace the estimates used during design. These are the numbers reported back to the modelling and control work, and they are the reason the parameter identification of Section 4.2 runs on measured rather than assumed values.

Table 28: As-built electrical measurements. Each replaces an estimate used earlier in the design.

Quantity	Measured	Replaces / feeds
Motor torque constant K_m	TBD	Table 6, datasheet value
Wheel inertia I_w (spin-down)	TBD	Table 6, CAD value
Viscous and Coulomb friction	TBD	C_b, C_w estimates
Command-to-encoder latency	TBD	Loop timing, Sec. 5.5.2
Achievable wheel speed on bus	TBD	$\omega_{\max}$, Sec. 4.3.1
Logic rail load, worst case	TBD	Table **??**
Runtime per charge under balance load	TBD	Demonstration planning